

# Capstone Project on Healthcare Data

Apr B Batch

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## Business Problem:

We all know that Health care is very important domain in the market. It is directly linked with the life of the individual; hence we have to be always be proactive in this particular domain. Money plays a major role in this domain, because sometime treatment becomes super costly and if any individual is not covered under the insurance then it will become a pretty tough financial situation for that individual. The companies in the medical insurance also want to reduce their risk by optimizing the insurance cost, because we all know a healthy body is in the hand of the individual only. If individual eat healthy and do proper exercise the chance of getting ill is drastically reduced.

## Goal & Objective:

The objective of this exercise is to build a model, using data that provide the optimum insurance cost for an individual. You have to use the health and habit related parameters for the estimated cost of insurance.

## Data dictionary:

- **applicant\_id:** Applicant unique ID
- **years\_of\_insurance\_with\_us:** Since how many years customer is taking policy from the same company only
- **regular\_checkup\_lasy\_year:** Number of times customers has done the regular health check up in last one year
- **adventure\_sports:** Customer is involved with adventure sports like climbing, diving etc.
- **Occupation:** Occupation of the customer.
- **visited\_doctor\_last\_1\_year:** Number of times customer has visited doctor in last one year
- **cholesterol\_level:** Cholesterol level of the customers while applying for insurance
- **daily\_avg\_steps:** Average daily steps walked by customers age Age of the customer
- **heart\_decs\_history:** Any past heart diseases
- **other\_major\_decs\_history:** Any past major diseases apart from heart like any operation
- **Gender:** Gender of the customer
- **avg\_glucose\_level:** Average glucose level of the customer while applying the insurance
- **bmi:** BMI of the customer while applying the insurance

- **smoking\_status:** Smoking status of the customer
- **Year\_last\_admitted:** When customer have been admitted in the hospital last time
- **Location:** Location of the hospital
- **weight:** Weight of the customer
- **covered\_by\_any\_other\_co:** mpany Customer is covered from any other insurance company
- **Alcohol:** Alcohol consumption status of the customer
- **exercise:** Regular exercise status of the customer
- **weight\_change\_in\_last\_one\_year:** How much variation has been seen in the weight of the customer in last year
- **fat\_percentage:** Fat percentage of the customer while applying the insurance
- **insurance\_cost:** Total Insurance cost

## Data Information:

```
Data columns (total 24 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   applicant_id                               25000 non-null   int64
1   years_of_insurance_with_us                 25000 non-null   int64
2   regular_checkup_lasy_year                  25000 non-null   int64
3   adventure_sports                           25000 non-null   int64
4   Occupation                                 25000 non-null   object
5   visited_doctor_last_1_year                 25000 non-null   int64
6   cholesterol_level                          25000 non-null   object
7   daily_avg_steps                           25000 non-null   int64
8   age                                         25000 non-null   int64
9   heart_decs_history                         25000 non-null   int64
10  other_major_decs_history                   25000 non-null   int64
11  Gender                                     25000 non-null   object
12  avg_glucose_level                         25000 non-null   int64
13  bmi                                        24010 non-null   float64
14  smoking_status                             25000 non-null   object
15  Year_last_admitted                         13119 non-null   float64
16  Location                                   25000 non-null   object
17  weight                                     25000 non-null   int64
18  covered_by_any_other_company               25000 non-null   object
19  Alcohol                                    25000 non-null   object
20  exercise                                   25000 non-null   object
21  weight_change_in_last_one_year             25000 non-null   int64
22  fat_percentage                            25000 non-null   int64
23  insurance_cost                             25000 non-null   int64
dtypes: float64(2), int64(14), object(8)
memory usage: 4.6+ MB
```

Figure 1: Data Info

- The data has in total 24 columns and 25000 rows.
- The dataset consists of 8 categorical columns and 16 numerical columns.
- Missing values are present in the bmi and Year\_last\_admitted columns.
- Converted object-type columns to the category data type helps reduce memory usage. Which has decreased from 4.6+ MB to 3.2 MB, this technique is generally useful for bigger datasets.
- There are no duplicate values in the dataset.

- The bmi column has 990 missing values, while the Year\_last\_admitted column contains 11,881 missing values in the dataset.

|                                | count        | mean         | std          | min         | 25%          | 50%          | 75%          | max          |
|--------------------------------|--------------|--------------|--------------|-------------|--------------|--------------|--------------|--------------|
| applicant_id                   | 25000.000000 | 17499.500000 | 7217.022701  | 5000.000000 | 11249.750000 | 17499.500000 | 23749.250000 | 29999.000000 |
| years_of_insurance_with_us     | 25000.000000 | 4.089040     | 2.606612     | 0.000000    | 2.000000     | 4.000000     | 6.000000     | 8.000000     |
| regular_checkup_last_year      | 25000.000000 | 0.773680     | 1.199449     | 0.000000    | 0.000000     | 0.000000     | 1.000000     | 5.000000     |
| adventure_sports               | 25000.000000 | 0.081720     | 0.273943     | 0.000000    | 0.000000     | 0.000000     | 0.000000     | 1.000000     |
| visited_doctor_last_1_year     | 25000.000000 | 3.104200     | 1.141663     | 0.000000    | 2.000000     | 3.000000     | 4.000000     | 12.000000    |
| daily_avg_steps                | 25000.000000 | 5215.889320  | 1053.179748  | 2034.000000 | 4543.000000  | 5089.000000  | 5730.000000  | 11255.000000 |
| age                            | 25000.000000 | 44.918320    | 16.107492    | 16.000000   | 31.000000    | 45.000000    | 59.000000    | 74.000000    |
| heart_decs_history             | 25000.000000 | 0.054640     | 0.227281     | 0.000000    | 0.000000     | 0.000000     | 0.000000     | 1.000000     |
| other_major_decs_history       | 25000.000000 | 0.098160     | 0.297537     | 0.000000    | 0.000000     | 0.000000     | 0.000000     | 1.000000     |
| avg_glucose_level              | 25000.000000 | 167.530000   | 62.729712    | 57.000000   | 113.000000   | 168.000000   | 222.000000   | 277.000000   |
| bmi                            | 24010.000000 | 31.393328    | 7.876535     | 12.300000   | 26.100000    | 30.500000    | 35.600000    | 100.600000   |
| Year_last_admitted             | 13119.000000 | 2003.892217  | 7.581521     | 1990.000000 | 1997.000000  | 2004.000000  | 2010.000000  | 2018.000000  |
| weight                         | 25000.000000 | 71.610480    | 9.325183     | 52.000000   | 64.000000    | 72.000000    | 78.000000    | 96.000000    |
| weight_change_in_last_one_year | 25000.000000 | 2.517960     | 1.690335     | 0.000000    | 1.000000     | 3.000000     | 4.000000     | 6.000000     |
| fat_percentage                 | 25000.000000 | 28.812280    | 8.632382     | 11.000000   | 21.000000    | 31.000000    | 36.000000    | 42.000000    |
| insurance_cost                 | 25000.000000 | 27147.407680 | 14323.691832 | 2468.000000 | 16042.000000 | 27148.000000 | 37020.000000 | 67870.000000 |

Figure 2: Statistical Summery

- applicant\_id is an ID variable and not useful for modelling. Hence, we have removed it.
- adventure\_sports, heart\_decs\_history, other\_major\_decs\_history column has 75% of the data to be zero.
- regular\_checkup\_last\_year column has 50% of the data to be zero.
- Visited\_doctor\_last\_1\_year, bmi, regular\_checkup\_last\_year, average\_daily\_steps seems to have outliers.

|                              | count | unique | top          | freq  |
|------------------------------|-------|--------|--------------|-------|
| Occupation                   | 25000 | 3      | Student      | 10169 |
| cholesterol_level            | 25000 | 5      | 150 to 175   | 8763  |
| Gender                       | 25000 | 2      | Male         | 16422 |
| smoking_status               | 25000 | 4      | never smoked | 9249  |
| Location                     | 25000 | 15     | Bangalore    | 1742  |
| covered_by_any_other_company | 25000 | 2      | N            | 17418 |
| Alcohol                      | 25000 | 3      | Rare         | 13752 |
| exercise                     | 25000 | 3      | Moderate     | 14638 |

Figure 3: Categorical Columns Description

- The most common region among individuals is **Bangalore**.
- The majority occupation is **Student**.
- There are more **male** individuals than **females**.
- The dataset includes **three unique occupations: Salaried, Student, and Business**.
- Cholesterol levels** fall into **five categories: 125-150, 150-175, 175-200, 200-225, and 225-250**.
- Gender** is categorized into **Male and Female**.

- **Smoking status** includes **four categories**: **Unknown, Formerly Smoked, Never Smoked, and Smokes.**
- Individuals belong to **15 different locations** in **India.**
- **Alcohol consumption** is classified into **three categories**: **Rare, Daily, and No.**
- **Exercise levels** are divided into **Moderate, Extreme, and No.**

Unique values in Occupation are :

Occupation

Student 10169

Business 10020

Salried 4811

Name: count, dtype: int64

\*\*\*\*\*

Unique values in cholesterol\_level are :

cholesterol\_level

150 to 175 8763

125 to 150 8339

200 to 225 2963

175 to 200 2881

225 to 250 2054

Name: count, dtype: int64

\*\*\*\*\*

Unique values in Gender are :

Gender

Male 16422

Female 8578

Name: count, dtype: int64

\*\*\*\*\*

Unique values in smoking\_status are :

smoking\_status

never smoked 9249

Unknown 7555

formerly smoked 4329

smokes 3867

Name: count, dtype: int64

\*\*\*\*\*

Unique values in Location are :

Location

Bangalore 1742

Jaipur 1706

Bhubaneswar 1704

Mangalore 1697

Delhi 1680

Ahmedabad 1677

Guwahati 1672

Chennai 1669

Kanpur 1664

Nagpur 1663

Mumbai 1658

Lucknow 1637

Pune 1622

```

Kolkata    1620
Surat      1589
Name: count, dtype: int64
*****

Unique values in covered_by_any_other_company are :
covered_by_any_other_company
N    17418
Y     7582
Name: count, dtype: int64
*****

Unique values in Alcohol are :
Alcohol
Rare    13752
No       8541
Daily   2707
Name: count, dtype: int64
*****

Unique values in exercise are :
exercise
Moderate 14638
Extreme   5248
No         5114
Name: count, dtype: int64
*****

```

## Data Imputation and Feature Engineering:

- Dropped column applicant\_id which is not adding any information.
- A new column (Years\_since\_last\_admission) has been added to indicate the number of years passed since the last admission (Years\_since\_last\_admission = current year - Year\_last\_admitted).
- There were missing values in column bmi (990) and Years\_since\_last\_admission (11881).
- Missing values in the BMI column were filled with the mean BMI based on gender.
- Missing values in Years\_since\_last\_admission were imputed with -1.
- The following columns were dropped: 'Location', and 'Year\_last\_admitted' as they don't contribute any relation with the dependent variable.
- Oversampling is not required during modeling as this is a regression problem.
- In the given dataset, categorical variables were converted into numerical values to prepare the data for regression modeling. The following transformations were performed:

### Occupation Encoding

- 'Salried' → 1
- 'Business' → 2
- 'Student' → 0

### Cholesterol Level Encoding

- '125 to 150' → 1
- '150 to 175' → 2

- '175 to 200' → 3
- '200 to 225' → 4
- '225 to 250' → 5

### **Gender Encoding**

- 'Male' → 1
- 'Female' → 0

### **Smoking Status Encoding**

- 'Unknown' → 0
- 'never smoked' → 1
- 'formerly smoked' → 2
- 'smokes' → 3

### **Alcohol Consumption Encoding**

- 'No' → 0
- 'Rare' → 1
- 'Daily' → 2

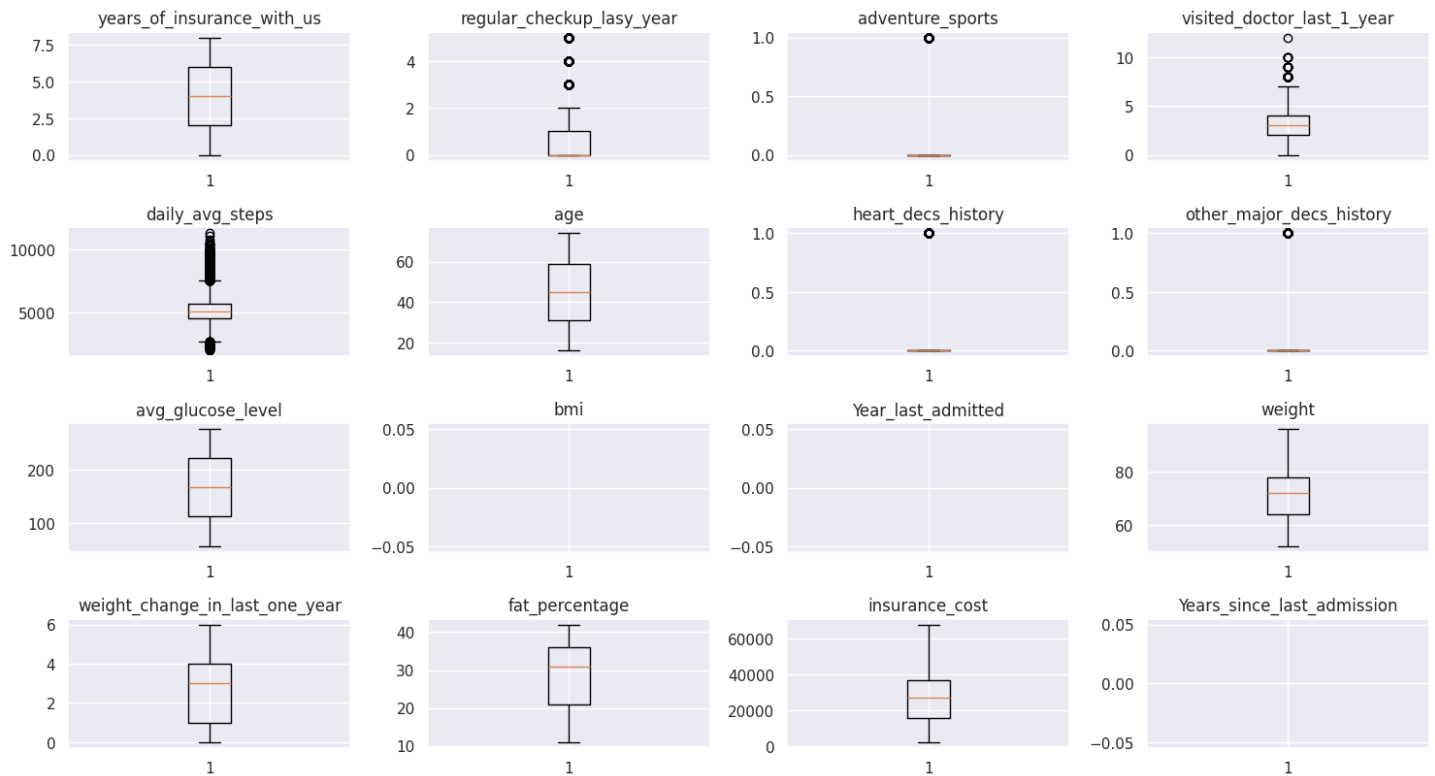
### **Exercise Category Encoding**

- 'No' → 0
- 'Moderate' → 1
- 'Extreme' → 2

### **Health Insurance Coverage by Another Company**

- 'N' (No) → 0
- 'Y' (Yes) → 1

## Outlier Treatment:



- There are much a few outliers in the data.
- However, we will not treat them as they are proper values.

# Exploratory Data Analysis (EDA) Summary:

## Univariate analysis:

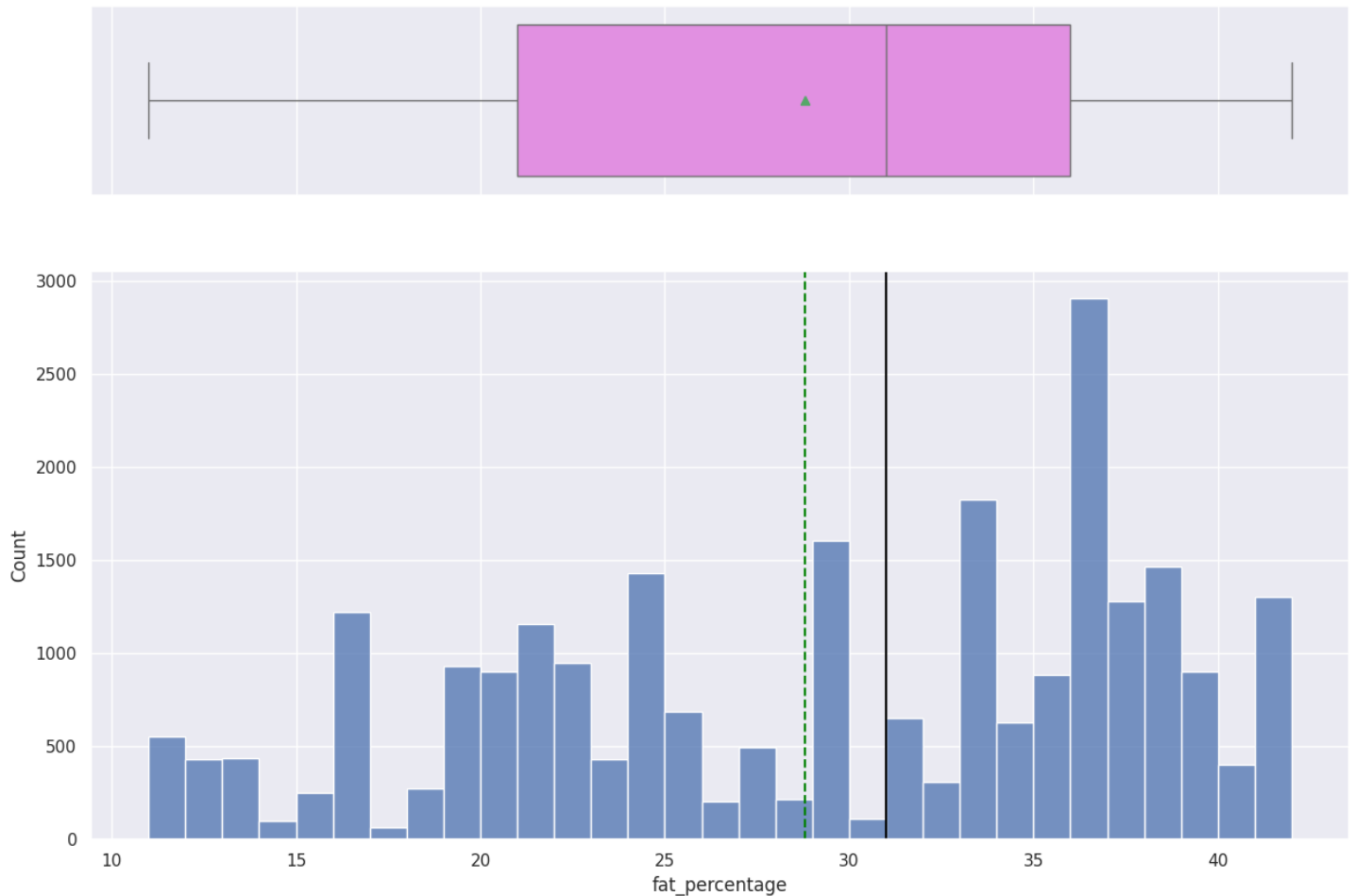


Figure 4: Univariate Fat Percentage

- The majority of the cohort has a high body fat percentage.
- The average body fat percentage in the cohort is 31.
- Most of the population (>50%) lies in between 33 to 40 percentile of the body weight which is an identifier of obesity.



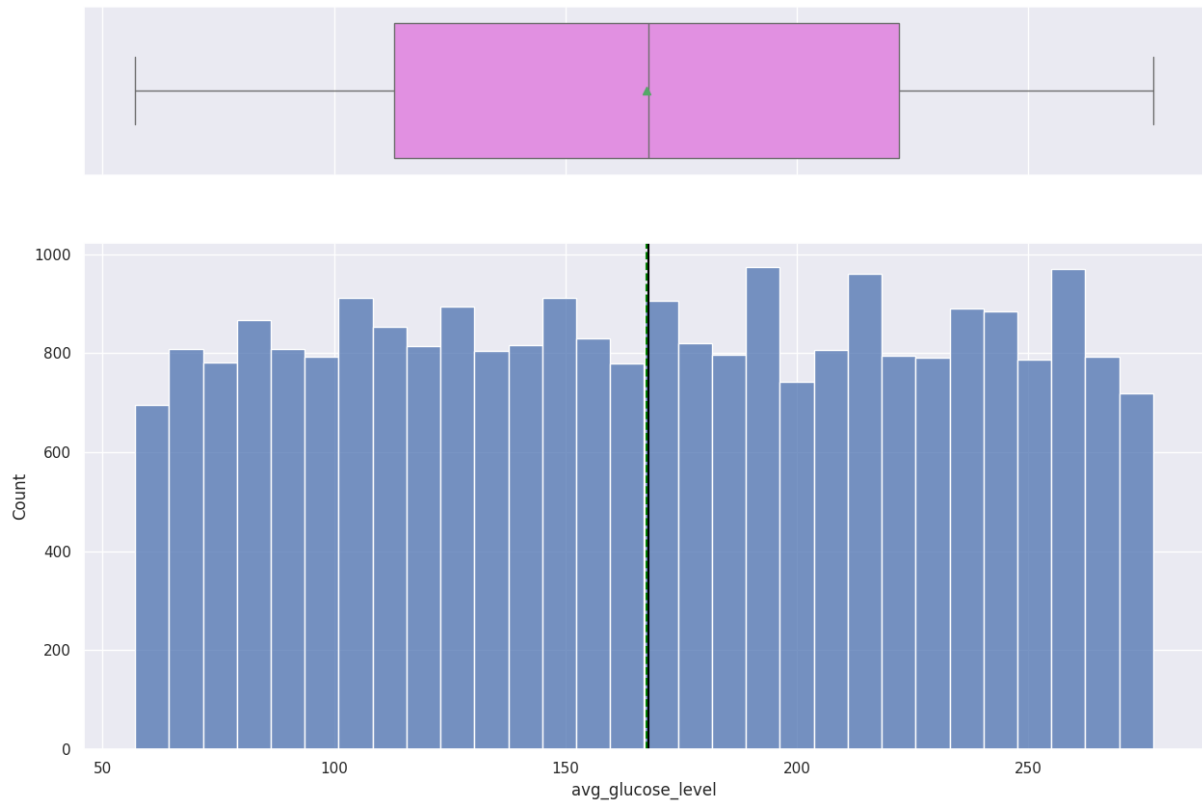


Figure 5: Univariate glucose level

- The distribution's mean and median glucose levels in the population are 168.
- Approximately 70% of the population is diabetic having blood glucose level more than 126 mg/dl.

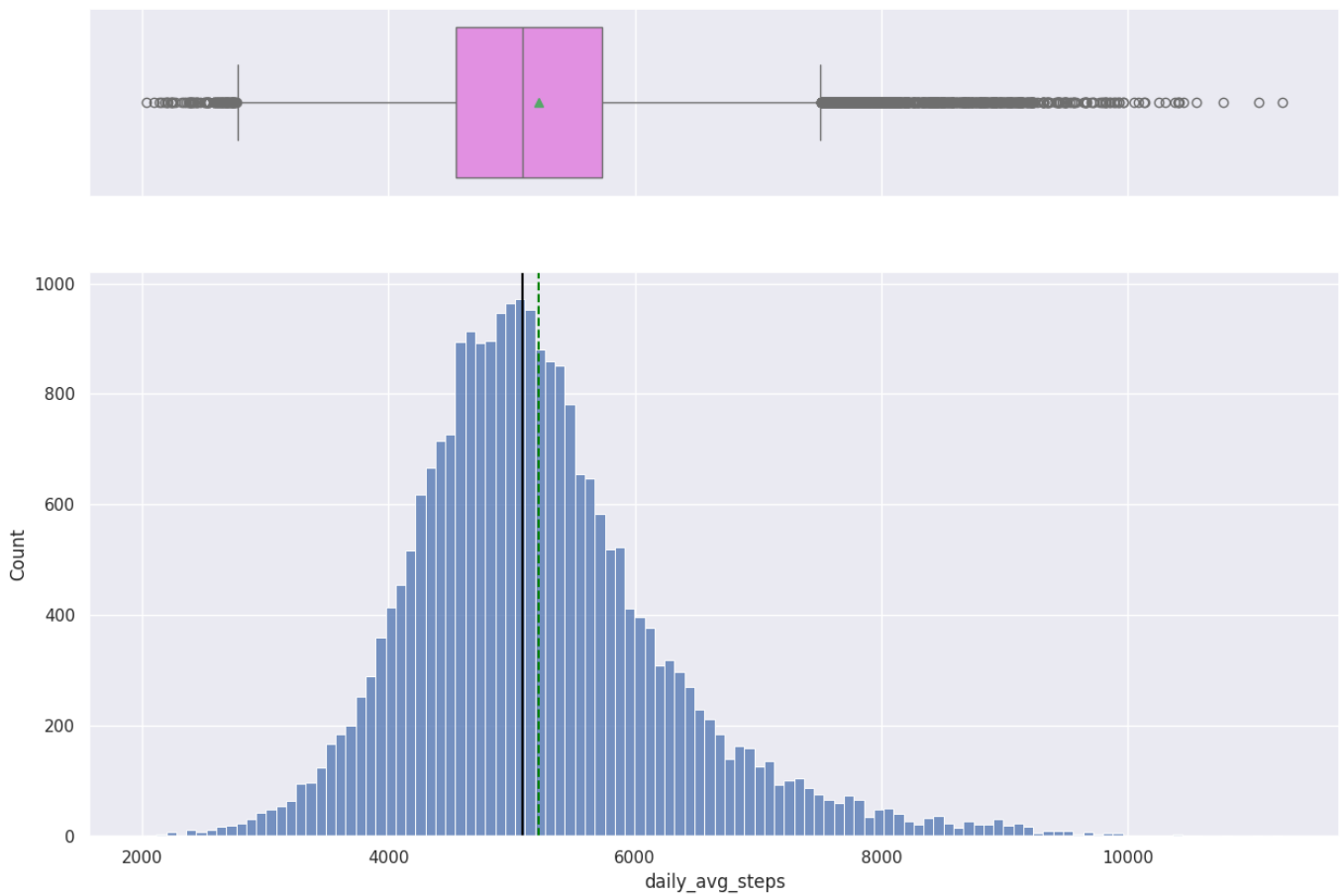


Figure 6: Univariate AVG steps

- The distribution appears to be **normal**, as the **mean and median** are nearly identical.
- Both **positive and negative outliers** are present.
- The majority of the population (approximately 75%) falls into the lightly or moderately active categories with average steps less than 6000 per day.

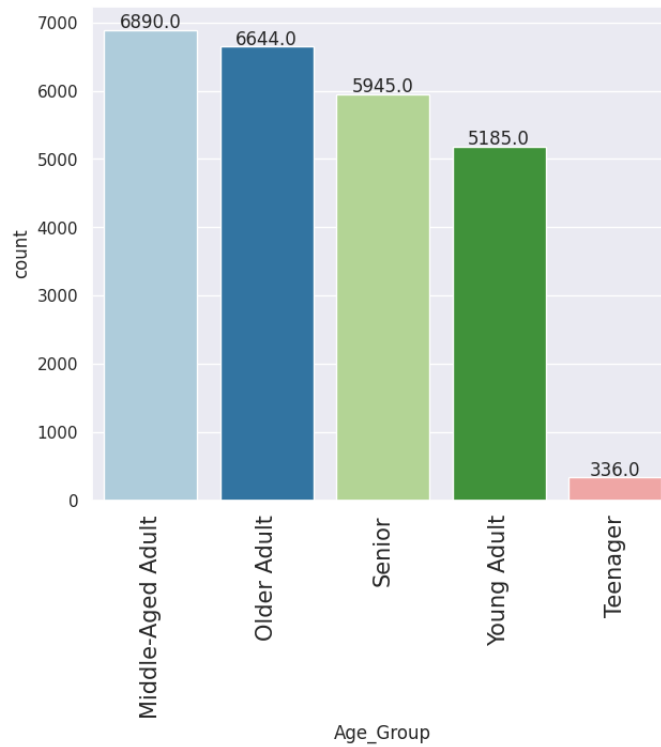


Figure 7: Age Categories

- There are very few teenage customers, while the majority of customers are middle-aged or elderly.

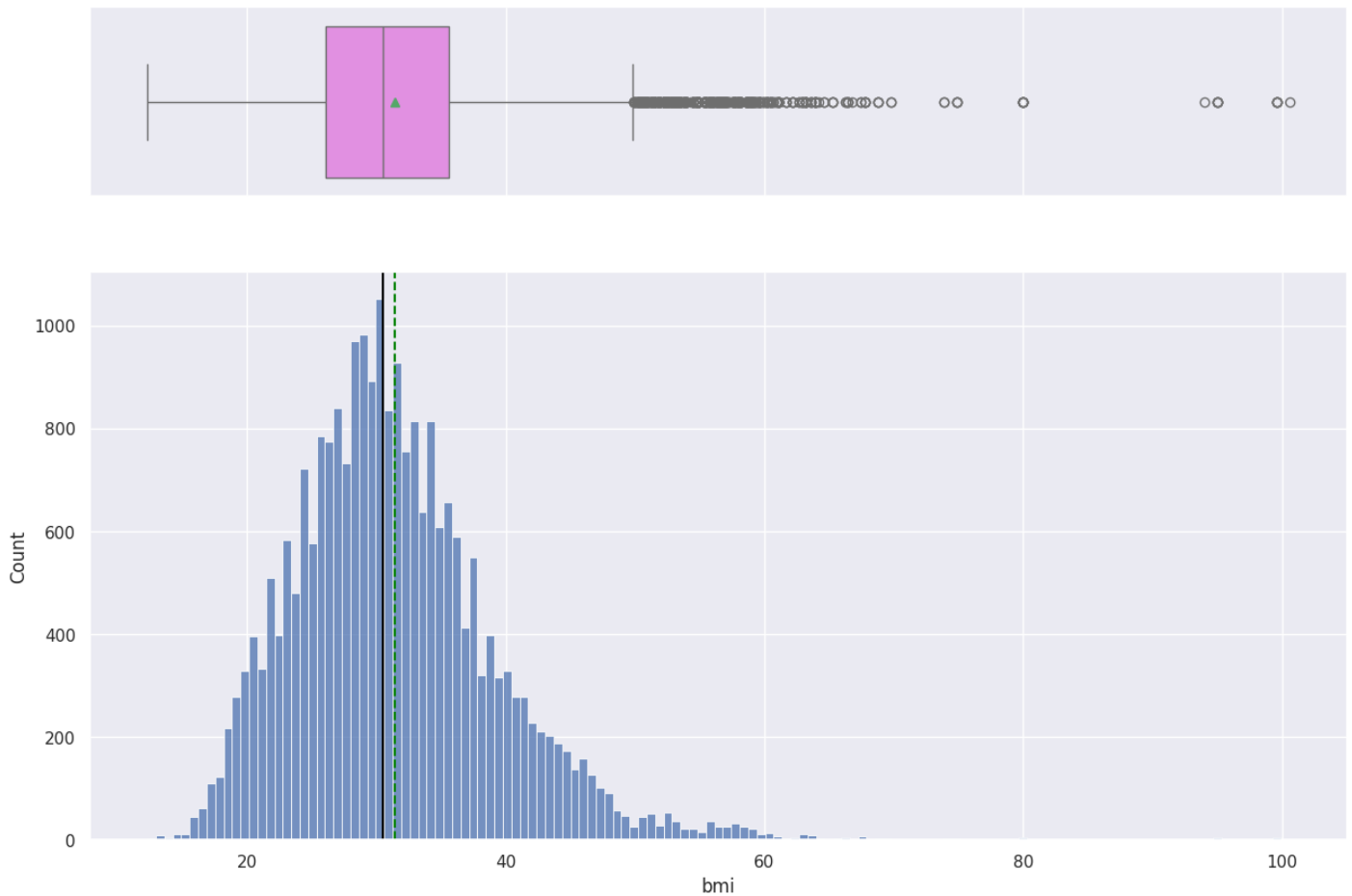
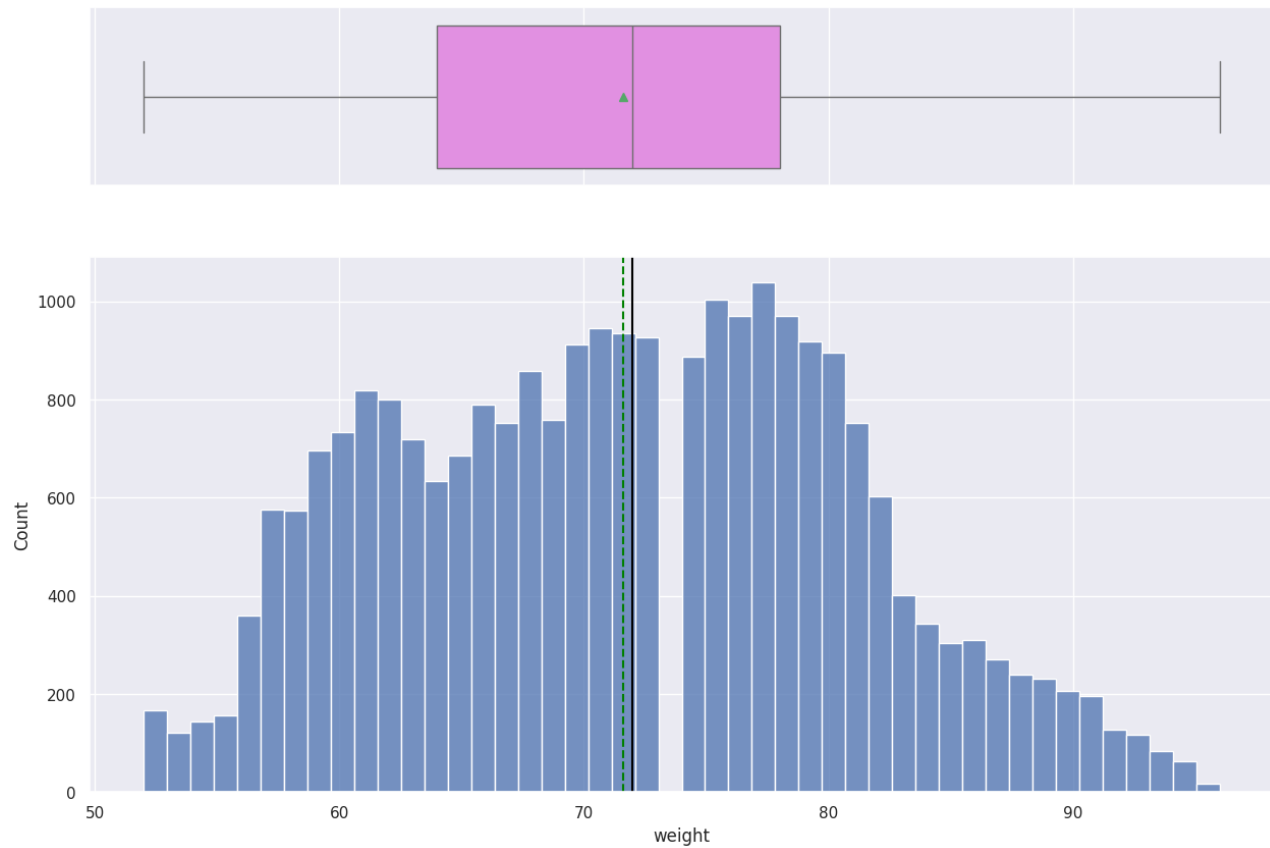


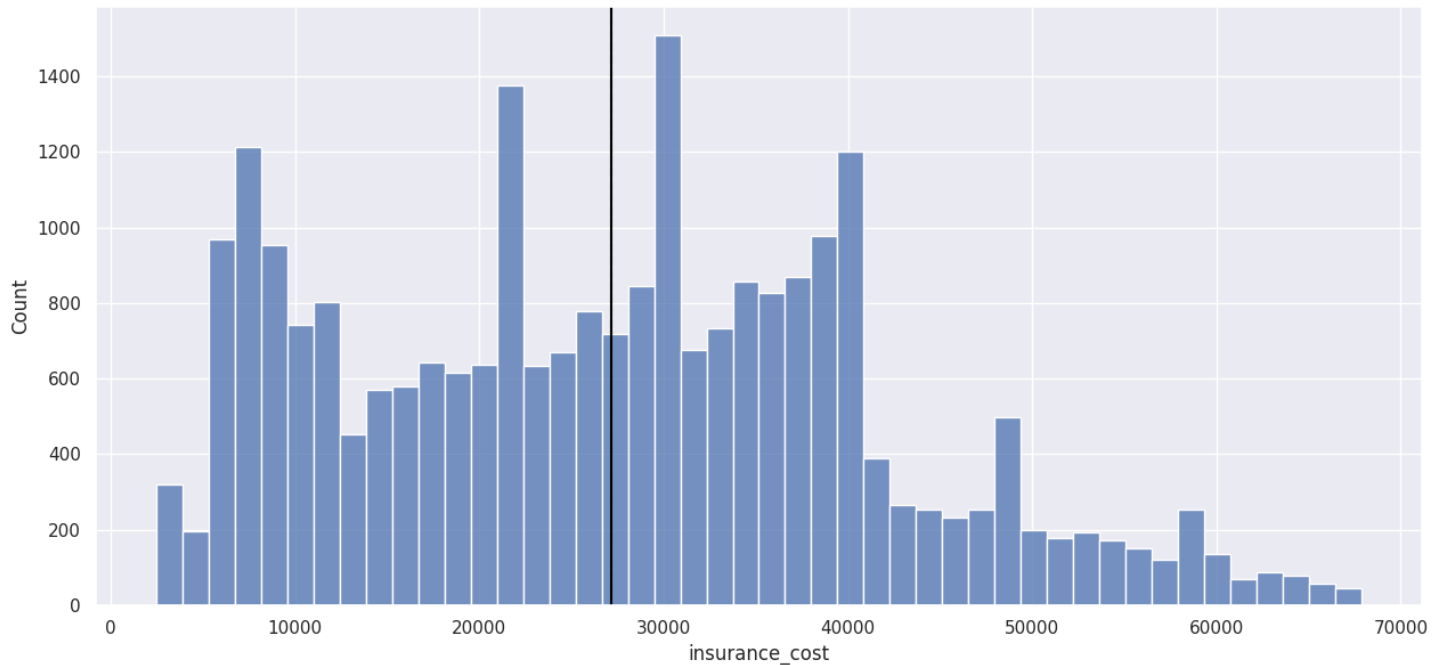
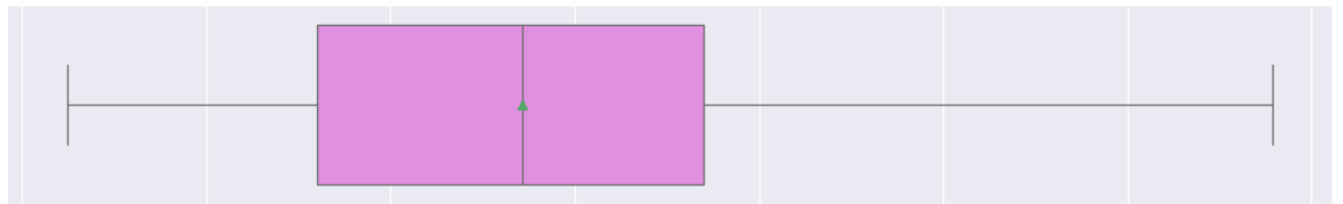
Figure 8: Univariate BMI

- The distribution is **normal** but **right-skewed**.
- Outliers are also present.
- Normal BMI range is between 18 to 24. However, a very small portion (about 25%) of the population has normal BMI range.



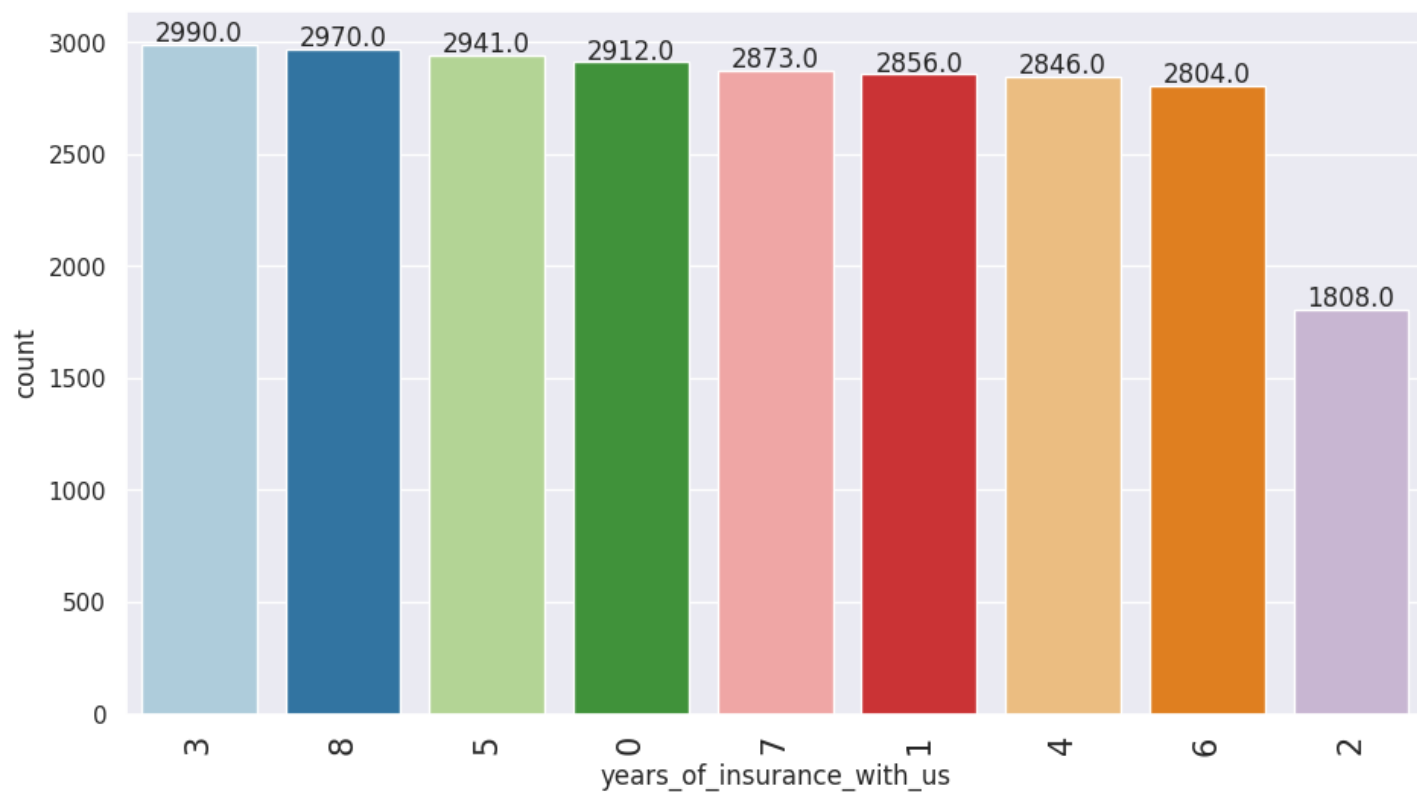
*Figure 9: Univariate Weight*

- The **mean and median** are nearly identical, indicating **no skewness** or **outliers** in the distribution.
- A significant portion (about 60%) of customers weighs more than 70 kg, which may imply overweight status.



*Figure 10: Univariate insurance cost*

- The **mean and median** for **insurance cost** are at the **same position**.
- A **positive skewness** is also observed.
- Distribution also suggests high insurance cost for majority of the population.



*Figure 11: Univariate Years of Insurance with the Company*

- Customers are **fairly evenly distributed** across **0 to 8 years** of engagement with the insurance company.

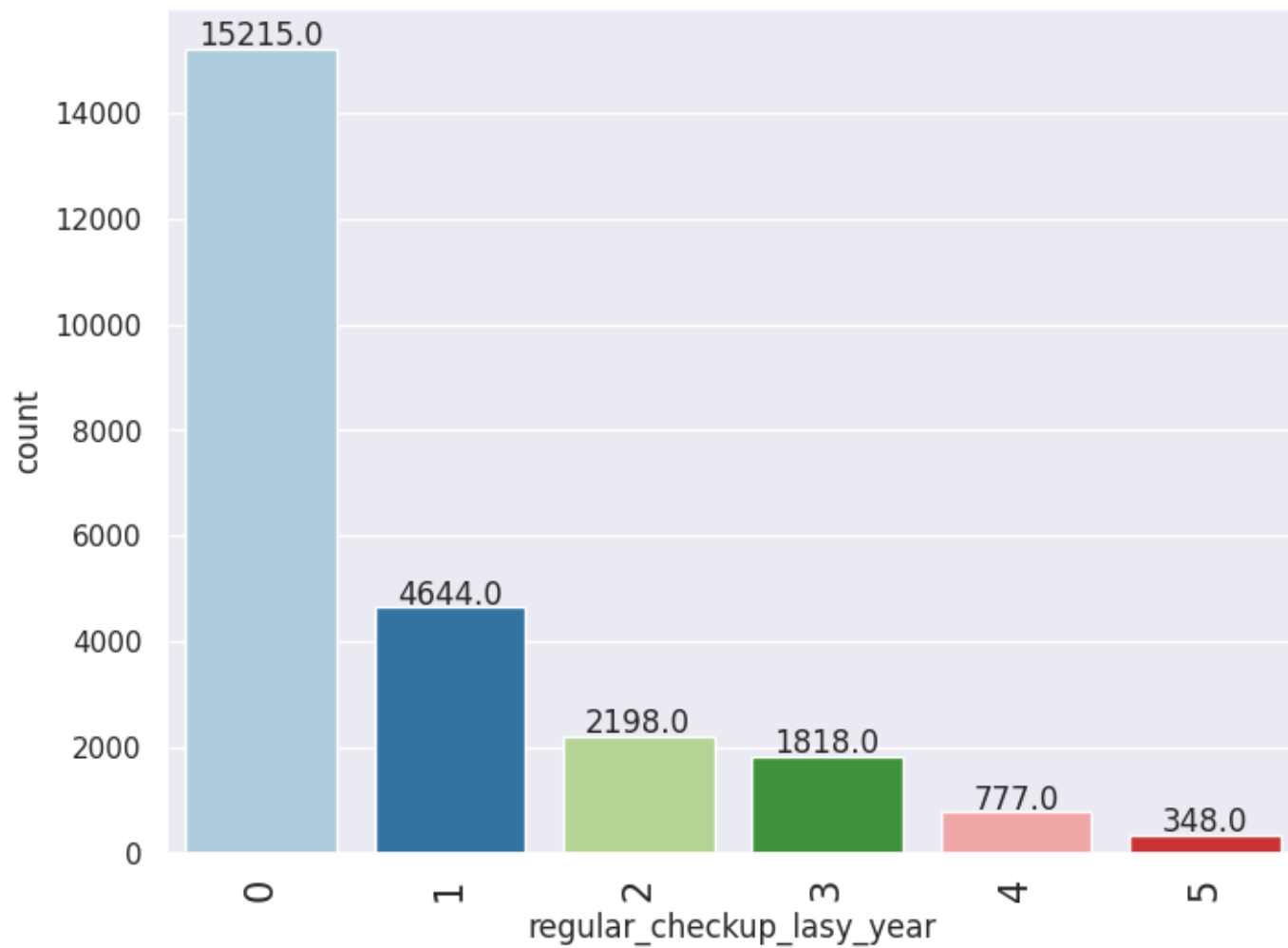


Figure 12: Univariate regular checkup last year

- The majority of patients did not have a regular checkup last year.

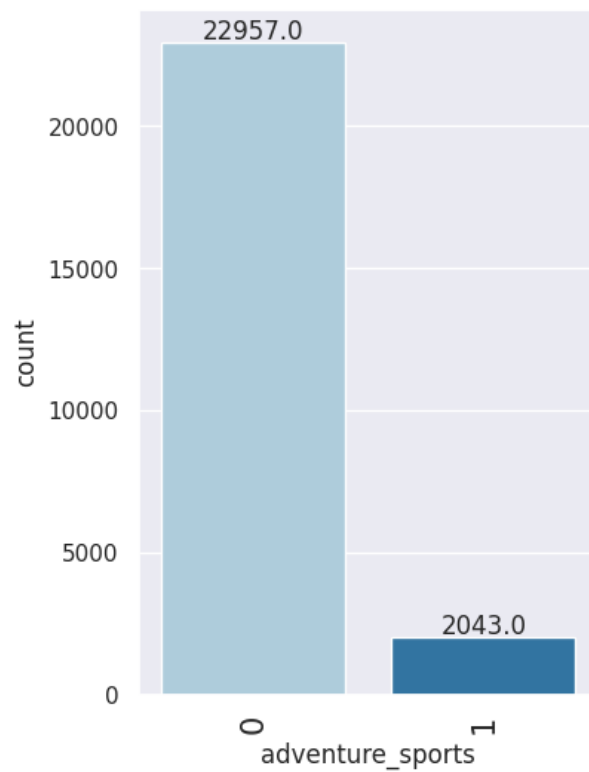


Figure 13: Univariate Adventure Sports

- Only a small number of people (2,043) participate in adventure sports like climbing and diving.

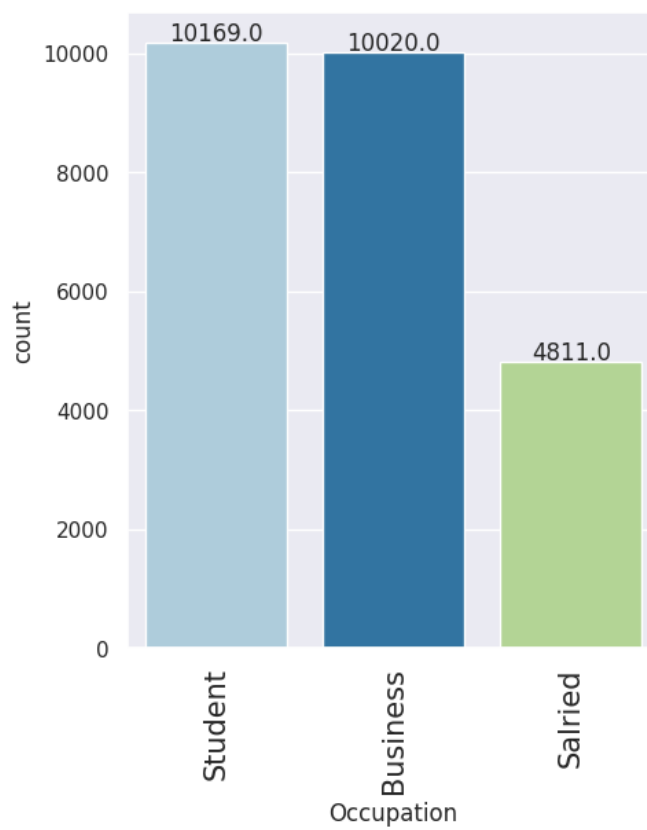


Figure 14: Univariate Occupation



- The majority of customers are engaged in **Student** or **Business** occupations, while **salaried employees** are relatively few.

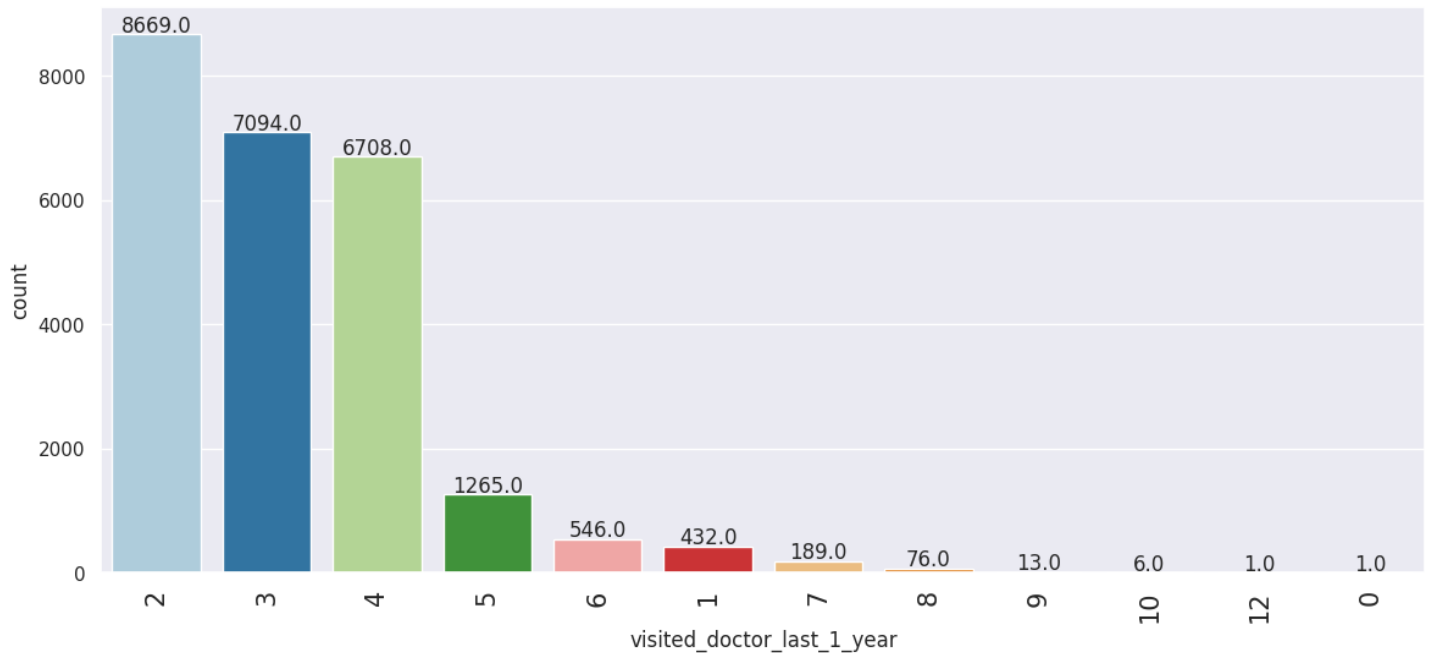


Figure 15: Univariate Doctor Visit Last Year

- Most patients visited the doctor around 2 to 5 times last year.

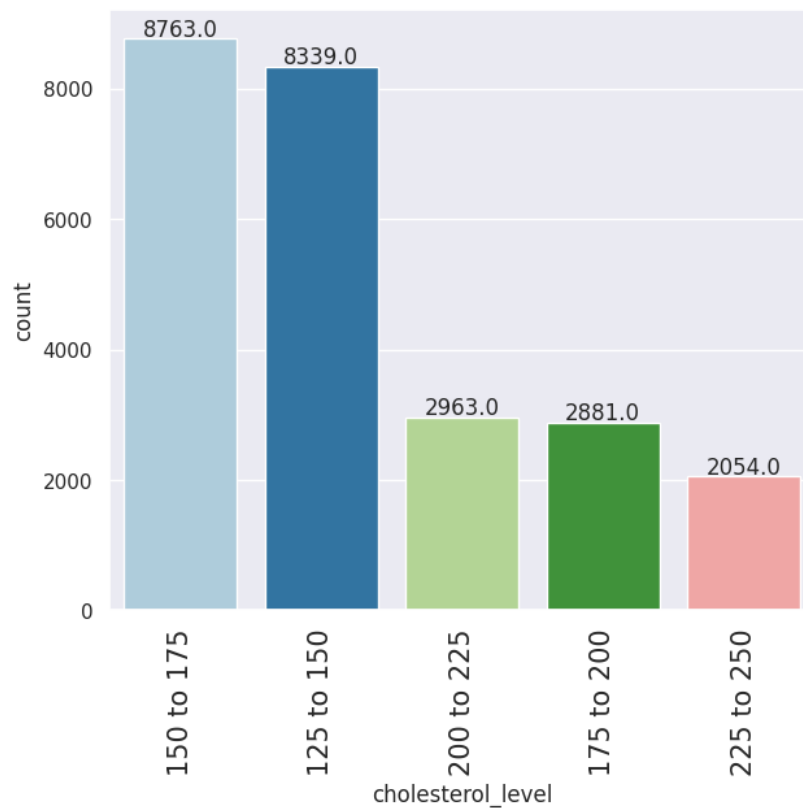


Figure 16: Univariate Cholesterol Level

- The majority of customers have a **cholesterol level** between **125-150** and **150-175**. Where normal cholesterol level is under 200 mg/dl.

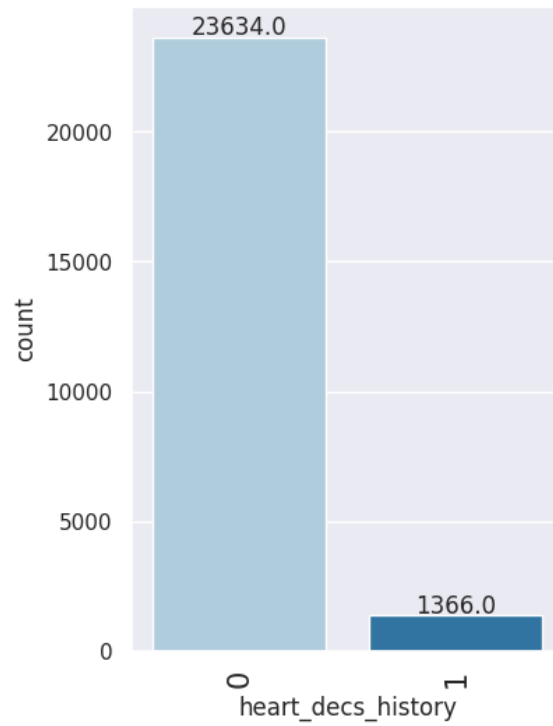


Figure 17: Univariate Heart Disease History

- Only a **small number of patients (1,366)** have a **history of heart disease**.

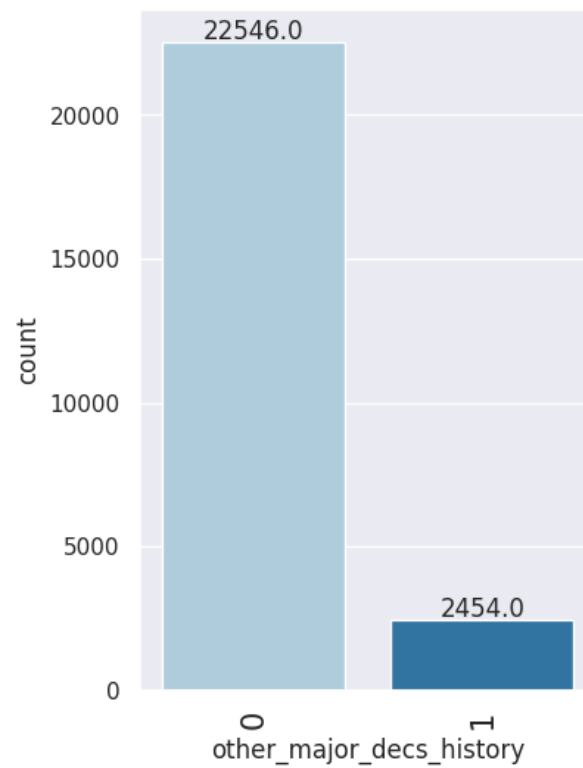


Figure 18: Univariate Other Disease History

- Only **2,454 people** have a history of **other major diseases**.

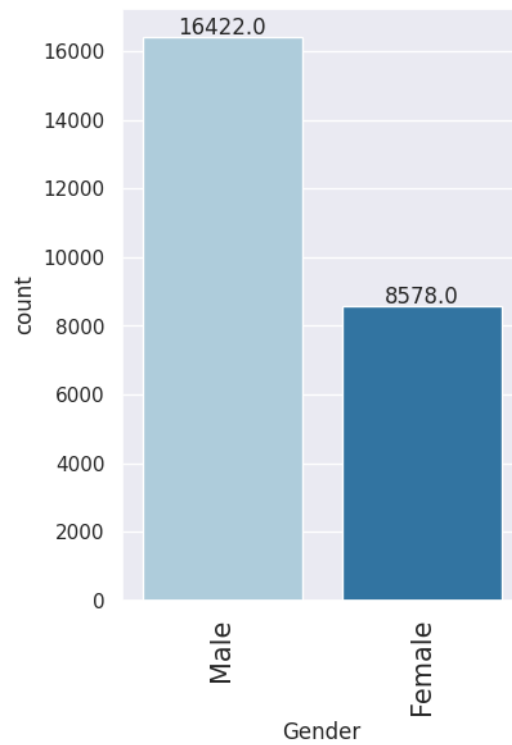


Figure 19: Univariate Gender

- The **gender distribution** in the data is approximately **2:1**.

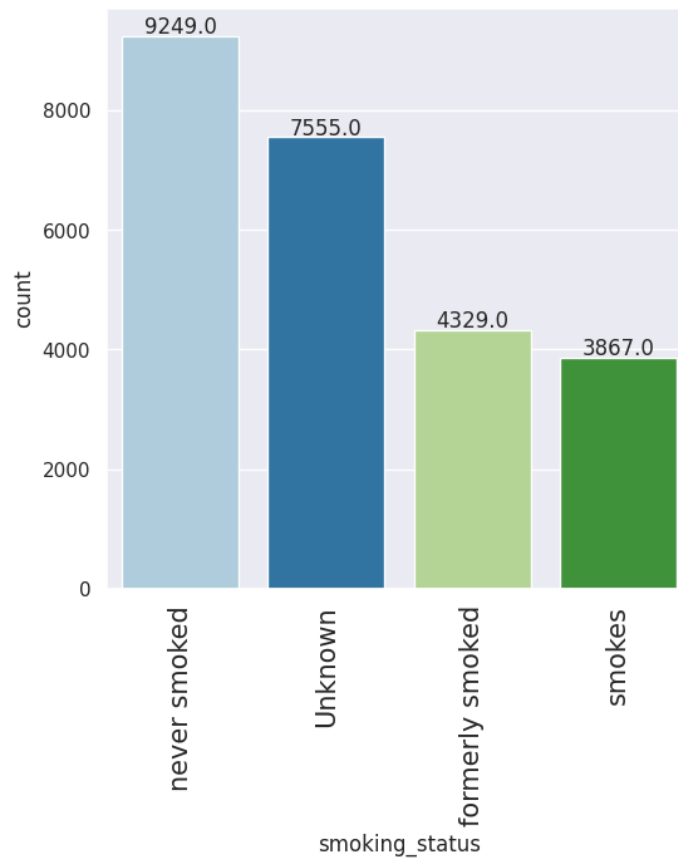


Figure 20: Univariate Smoking Status

- The majority of customers **never smoked** or have an **unknown smoking status**.

- The total number of customers who **formerly smoked** or **currently smoke** is approximately **8.2K**.

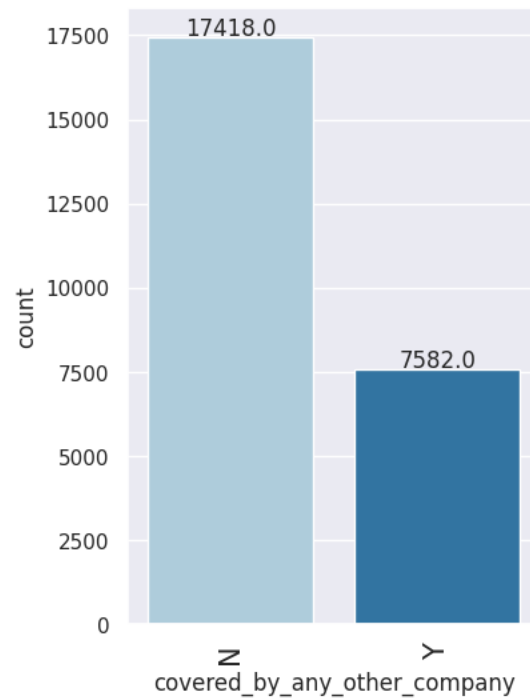


Figure 21: Univariate Insurance Covering by Other Company

- Around **7,582 customers** are covered by **another insurance**, accounting for **one-third** of the total dataset.

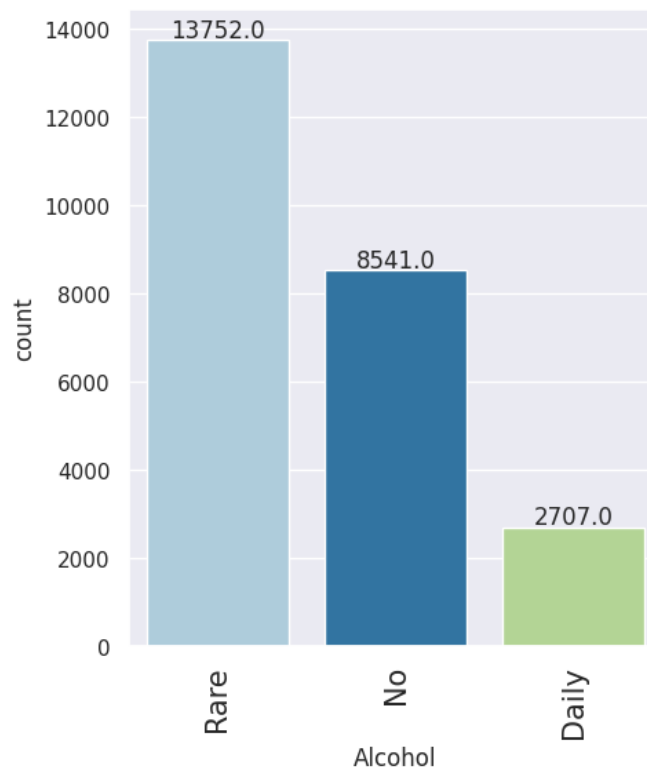


Figure 22: Univariate Alcohol

- Most of the customers have rarely or did not have any previous alcoholic condition.

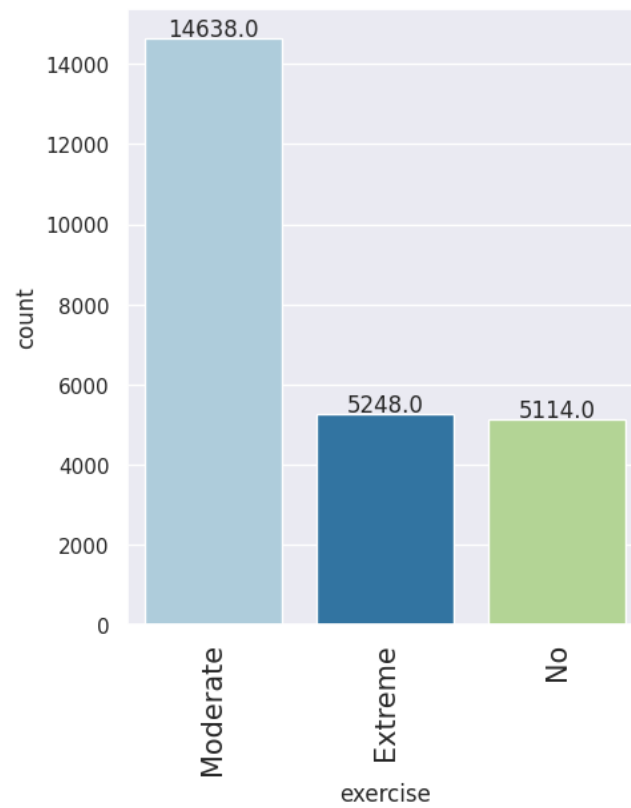


Figure 23: Univariate Exercise

- The majority of customers engage in moderate exercise.

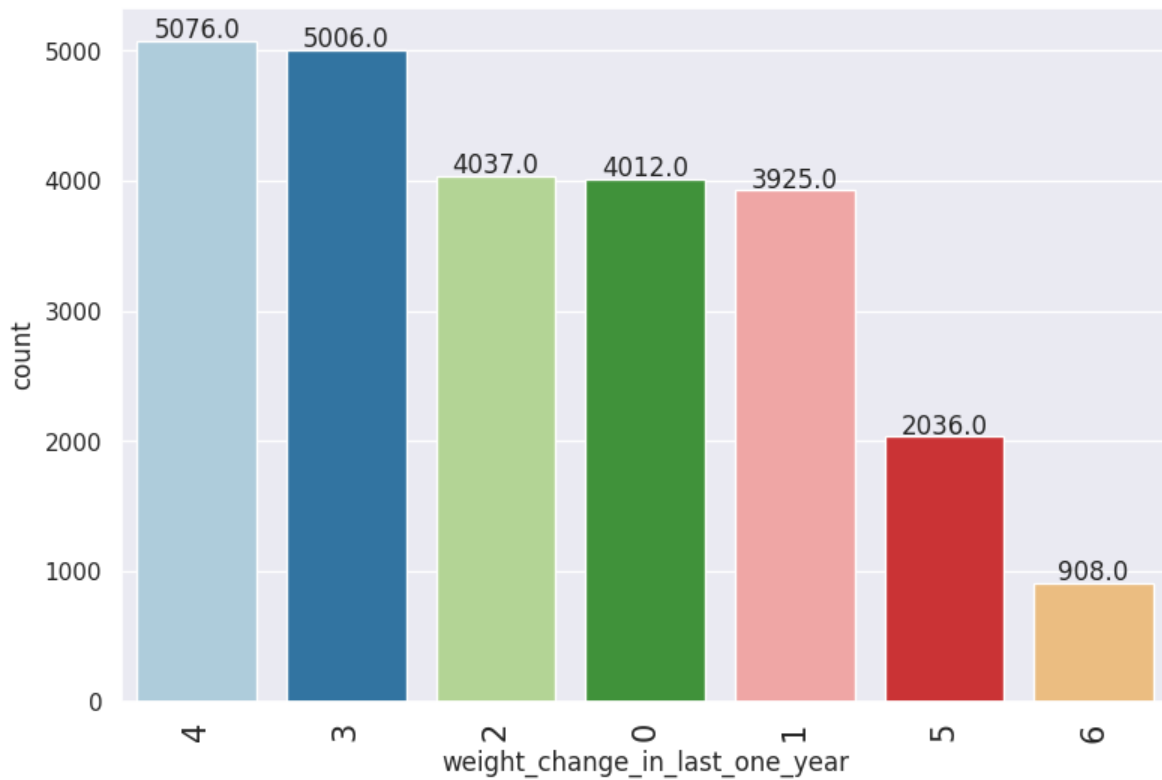


Figure 24: Univariate Weight Change in Last year

- Most weight changes in the last year ranged between 1 to 4 kg.

## Bivariate Analysis:

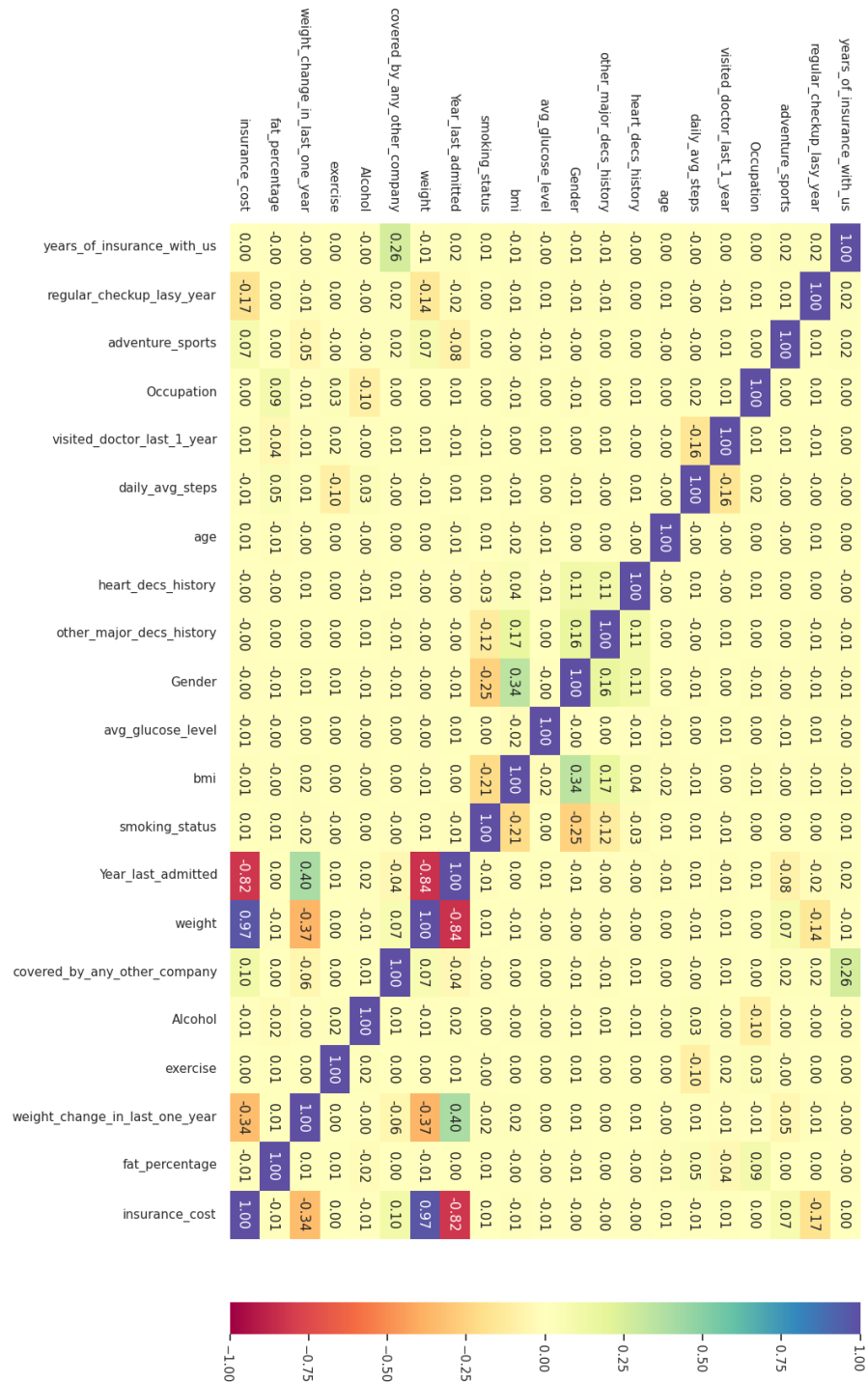


Figure 25: Correlation Plot

- A **strong positive correlation** is observed between **insurance cost** and **weight**.
- A **strong negative correlation** exists between **year last admitted** and both **insurance cost** and **weight**.

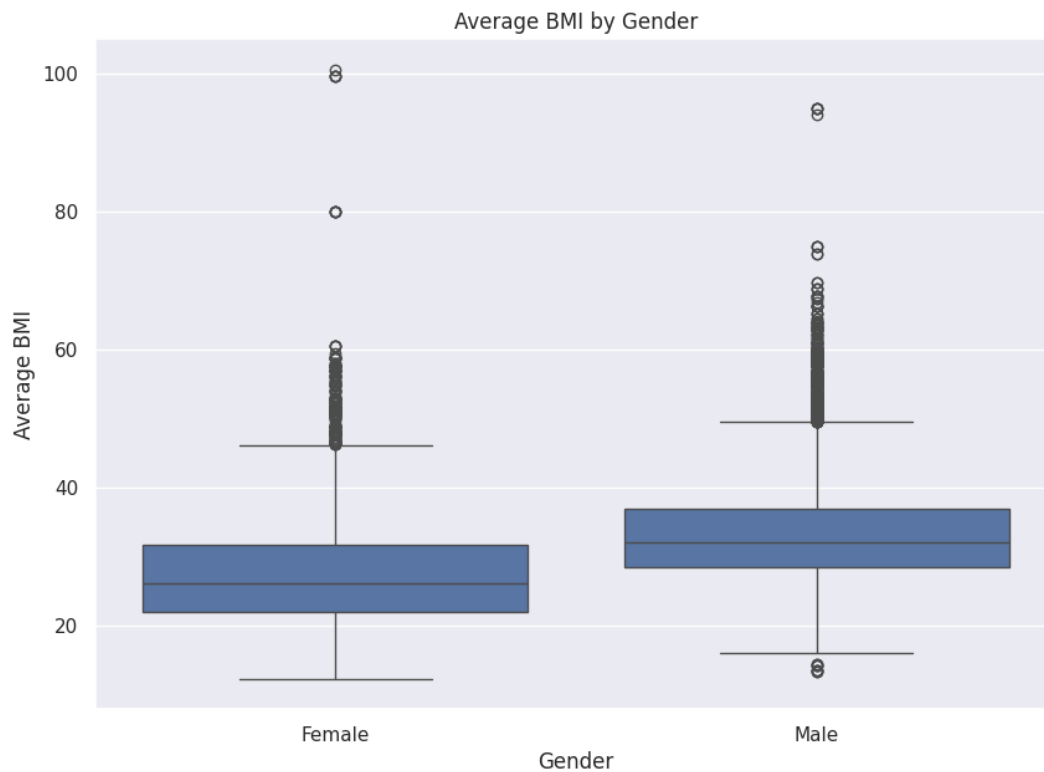


Figure 26: Avg BMI by Gender

- Female customers have a **lower average BMI** compared to male customers.

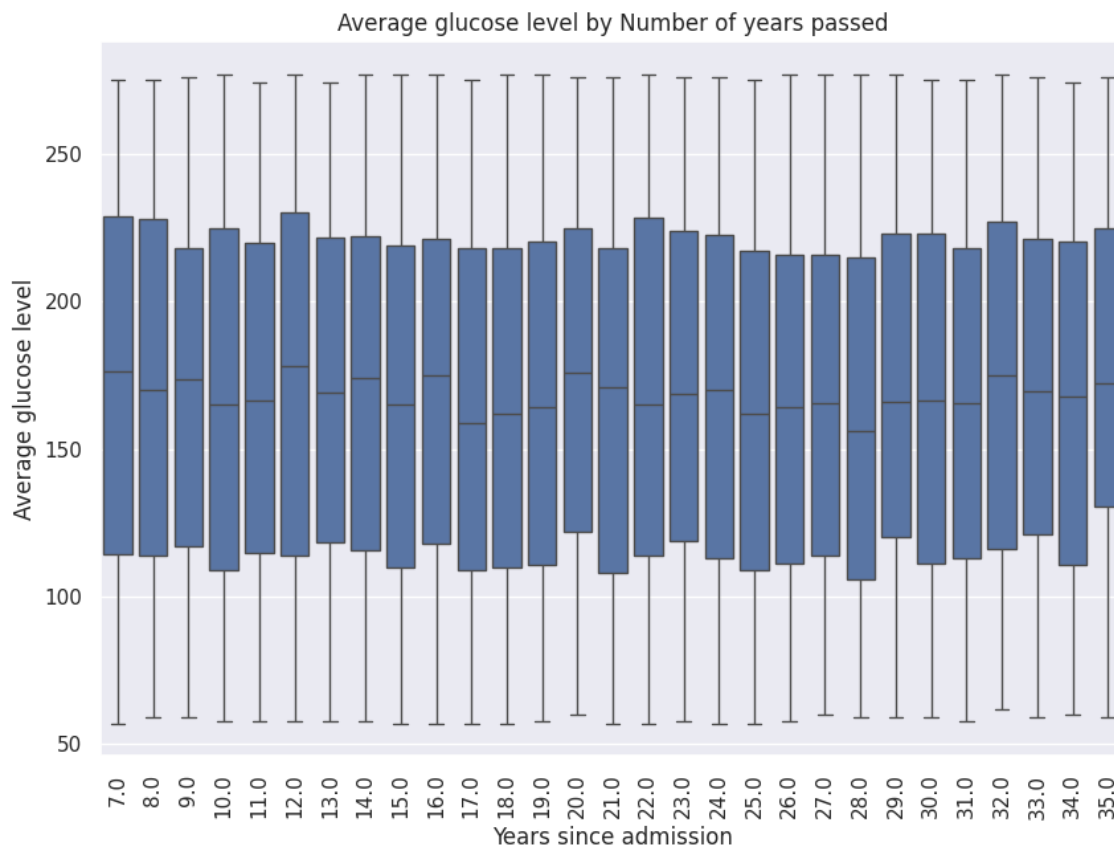


Figure 27: AVG Glucose Level by Number of Years Passed

- The **average glucose levels** have remained **relatively constant** over the years.

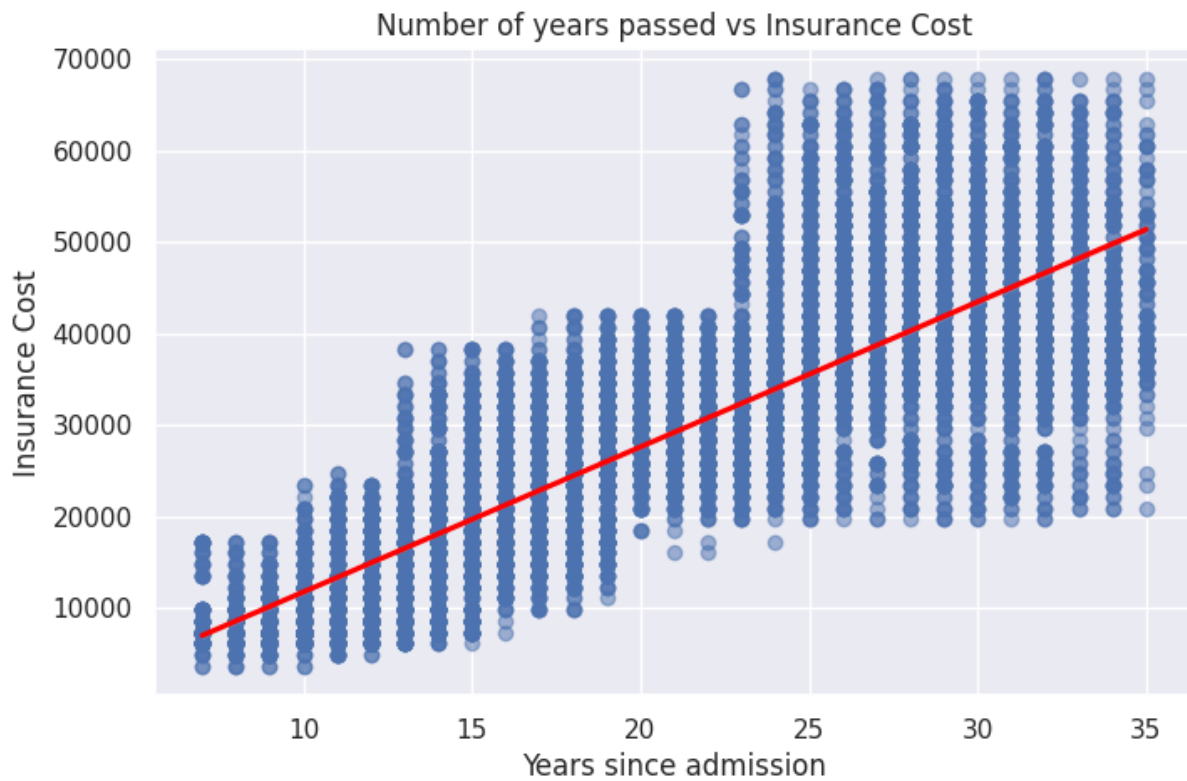


Figure 28: Number of years passed vs Insurance Cost

As the **number of years since last admission increases**, **insurance costs also increase**. This suggests that **insurance costs were higher in the past and have decreased over the years**.

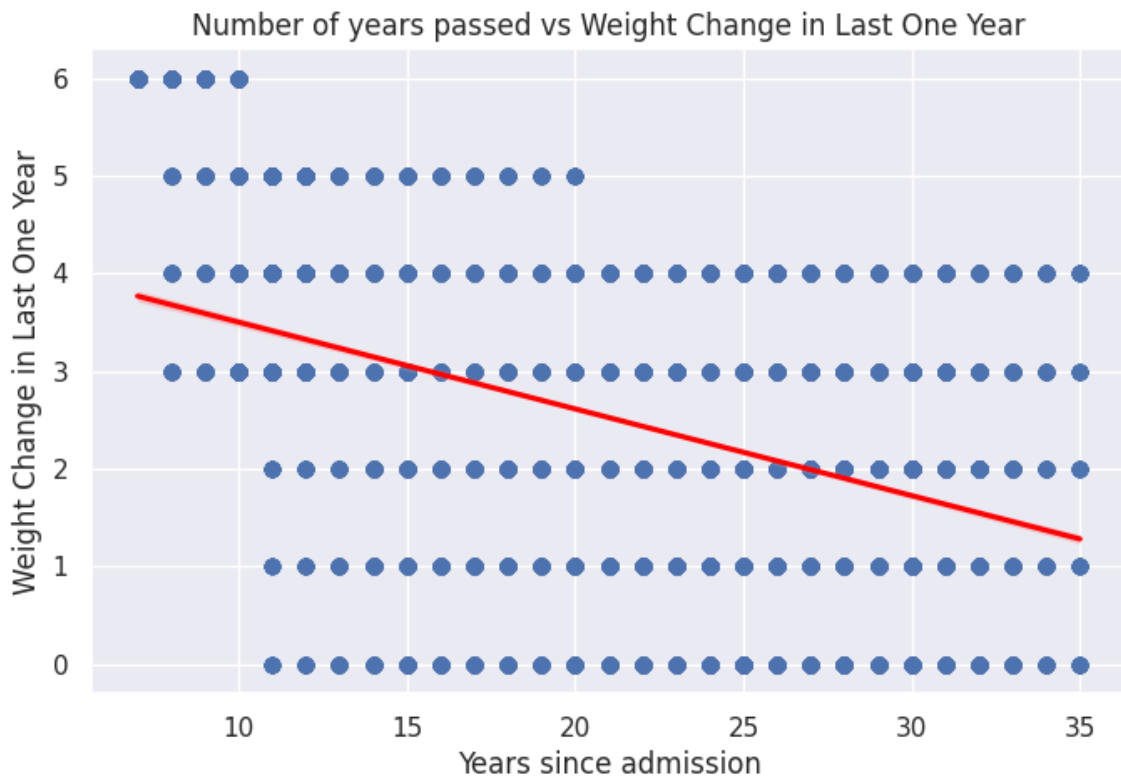


Figure 29: Number of years passed vs Weight Change in Last One Year



- The relationship between **number of years passed** and **weight change in the last year** shows a **negative correlation**. That means customers who were admitted a long time ago experience less weight fluctuation in the last year compared to those with more recent admissions.

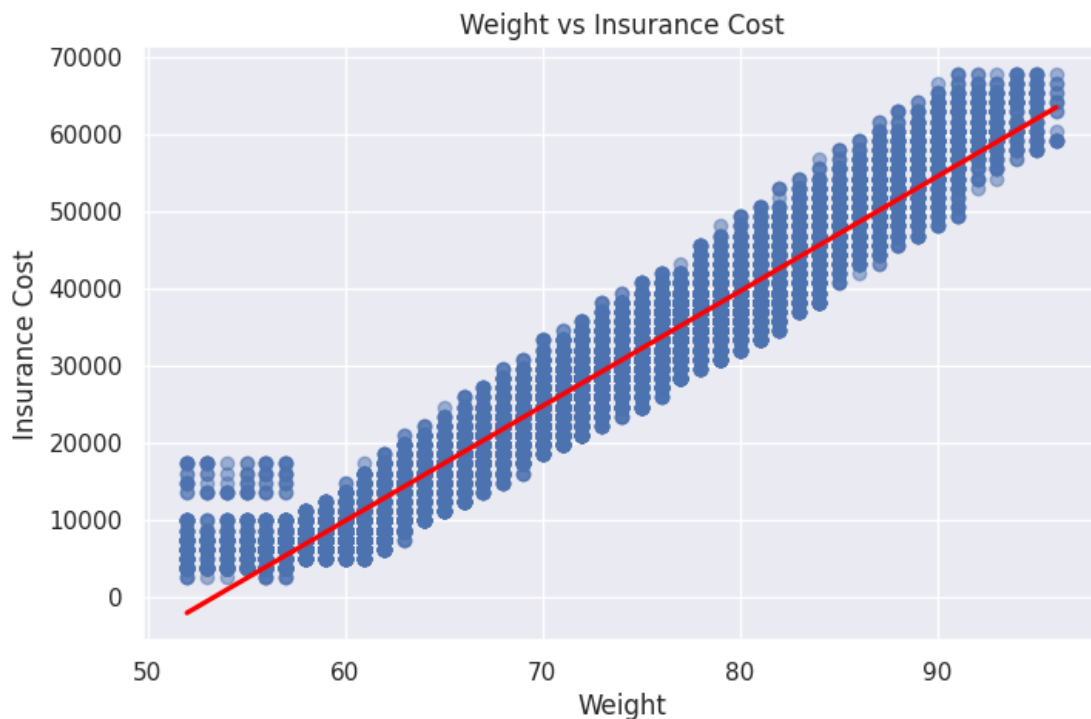


Figure 30: Customer Weight vs Insurance Cost

- As a customer's **weight increases**, their **insurance cost** also rises. High insurance cost has direct relation with the overweight.

## Modeling:

We have below columns in considerations for modeling:

'years\_of\_insurance\_with\_us', 'regular\_checkup\_lasy\_year', 'adventure\_sports', 'Occupation', 'visited\_doctor\_last\_1\_year', 'cholesterol\_level', 'daily\_avg\_steps', 'age', 'heart\_decs\_history', 'other\_major\_decs\_history', 'Gender', 'avg\_glucose\_level', 'bmi', 'smoking\_status', 'weight', 'covered\_by\_any\_other\_company', 'Alcohol', 'exercise', 'weight\_change\_in\_last\_one\_year', 'fat\_percentage', 'Years\_since\_last\_admission'

Splitting the data in 70:30 ratio for train to test data is done.

Number of rows in train data = 17500

Number of rows in test data = 7500

## Linear Regression Model:

```

=====
                        OLS Regression Results
=====
Dep. Variable:          insurance_cost    R-squared:                0.880
Model:                  OLS              Adj. R-squared:         0.880
Method:                 Least Squares    F-statistic:            6124.
Date:                   Thu, 06 Mar 2025  Prob (F-statistic):      0.00
Time:                   04:59:38         Log-Likelihood:         1040.6
No. Observations:      17500            AIC:                   -2037.
Df Residuals:          17478            BIC:                   -1866.
Df Model:              21
Covariance Type:       nonrobust
=====
                        coef    std err          t      P>|t|      [0.025    0.975]
-----
const                5.5643      0.088     63.533    0.000      5.393      5.736
years_of_insurance_with_us  0.0005      0.001      0.653    0.514     -0.001      0.002
regular_checkup_lasy_year -0.0132      0.002     -8.706    0.000     -0.016     -0.010
adventure_sports        0.0332      0.006      5.241    0.000      0.021      0.046
Occupation             -0.0048      0.002     -2.144    0.032     -0.009     -0.000
visited_doctor_last_1_year -0.0002      0.002     -0.117    0.907     -0.003      0.003
cholesterol_level       5.398e-05    0.002      0.034    0.973     -0.003      0.003
daily_avg_steps         -0.0022      0.009     -0.247    0.805     -0.020      0.015
age                    -1.612e-05    0.000     -0.151    0.880     -0.000      0.000
heart_decs_history      0.0054      0.008      0.693    0.488     -0.010      0.021
other_major_decs_history 0.0090      0.006      1.496    0.135     -0.003      0.021
Gender                 -0.0022      0.004     -0.531    0.595     -0.010      0.006
avg_glucose_level       0.0048      0.004      1.163    0.245     -0.003      0.013
bmi                    0.0022      0.008      0.261    0.794     -0.014      0.018
smoking_status          0.0005      0.002      0.284    0.777     -0.003      0.004
weight                 0.0629      0.000    282.637    0.000      0.063      0.063
covered_by_any_other_company 0.0463      0.004     11.852    0.000      0.039      0.054
Alcohol                0.0013      0.003      0.465    0.642     -0.004      0.007
exercise               0.0029      0.003      1.066    0.286     -0.002      0.008
weight_change_in_last_one_year -0.0299      0.001    -27.161    0.000     -0.032     -0.028
fat_percentage          0.0001      0.000      0.659    0.510     -0.000      0.001
Years_since_last_admission 0.0008      0.000      4.091    0.000      0.000      0.001
=====
Omnibus:              933.286    Durbin-Watson:           1.970
Prob(Omnibus):         0.000    Jarque-Bera (JB):        1283.208
Skew:                  -0.500    Prob(JB):                2.26e-279
Kurtosis:              3.871    Cond. No.                 4.67e+03
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.67e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

```

Figure 31: Initial Regression Model

We can see the model has high p values.

Series before feature selection:

```

const                2578.850824
years_of_insurance_with_us    1.506506
regular_checkup_lasy_year    1.108365
adventure_sports          1.008766
Occupation              1.372266
visited_doctor_last_1_year    1.035088
cholesterol_level        1.364573
daily_avg_steps          1.045616
age                    1.000710
heart_decs_history        1.025073
other_major_decs_history    1.050622
Gender                 1.256624
avg_glucose_level        1.001210
bmi                   1.227343
smoking_status          1.125678
weight                1.458831
covered_by_any_other_company    1.089420
Alcohol               1.013672
exercise              1.013997
weight_change_in_last_one_year    1.171785
fat_percentage          1.016674
Years_since_last_admission    1.837070
dtype: float64

```

Figure 32: Initial Model VIF Score

- The Variance Inflation Factor (VIF) scores are below 3, indicating that multicollinearity is not a concern in the dataset. This suggests that the independent variables are not highly correlated with each other, ensuring the stability and reliability of the regression model.
- Removed variables having higher p value.
- Columns with low p value 'const', 'regular\_checkup\_lasy\_year', 'adventure\_sports', 'Occupation', 'weight', 'covered\_by\_any\_other\_company', 'weight\_change\_in\_last\_one\_year', 'Years\_since\_last\_admission'.

```

=====
OLS Regression Results
=====
Dep. Variable:    insurance_cost    R-squared:        0.880
Model:            OLS              Adj. R-squared:    0.880
Method:           Least Squares     F-statistic:      1.838e+04
Date:             Thu, 06 Mar 2025   Prob (F-statistic): 0.00
Time:             04:59:38          Log-Likelihood:    1037.0
No. Observations: 17500            AIC:              -2058.
Df Residuals:     17492            BIC:              -1996.
Df Model:         7
Covariance Type:  nonrobust
=====

```

|                                | coef    | std err | t       | P> t  | [0.025 | 0.975] |
|--------------------------------|---------|---------|---------|-------|--------|--------|
| const                          | 5.5872  | 0.017   | 337.848 | 0.000 | 5.555  | 5.620  |
| regular_checkup_lasy_year      | -0.0130 | 0.001   | -8.694  | 0.000 | -0.016 | -0.010 |
| adventure_sports               | 0.0332  | 0.006   | 5.237   | 0.000 | 0.021  | 0.046  |
| Occupation                     | -0.0047 | 0.002   | -2.438  | 0.015 | -0.008 | -0.001 |
| weight                         | 0.0629  | 0.000   | 291.372 | 0.000 | 0.062  | 0.063  |
| covered_by_any_other_company   | 0.0468  | 0.004   | 12.173  | 0.000 | 0.039  | 0.054  |
| weight_change_in_last_one_year | -0.0299 | 0.001   | -27.163 | 0.000 | -0.032 | -0.028 |
| Years_since_last_admission     | 0.0008  | 0.000   | 5.238   | 0.000 | 0.001  | 0.001  |

```

=====
Omnibus:          932.764    Durbin-Watson:      1.970
Prob(Omnibus):    0.000    Jarque-Bera (JB):    1284.243
Skew:             -0.500    Prob(JB):            1.35e-279
Kurtosis:         3.873    Cond. No.            702.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

## Model Performance:

- Training Set Metrics: {'R<sup>2</sup>': 0.8803, 'Adj. R<sup>2</sup>': 0.8803, 'RMSE': 0.23, 'MAE': 0.18, 'MAPE (%)': 1.82}
- Test Set Metrics: {'R<sup>2</sup>': 0.8832, 'Adj. R<sup>2</sup>': 0.883, 'RMSE': 0.22, 'MAE': 0.17, 'MAPE (%)': 1.78}

## Observations:

- **Good Model Fit:**
  - The **R<sup>2</sup> values** are 0.8803 (training) and 0.8832 (test), indicating that the model explains around **88%** of the variance in the target variable.
  - **Adjusted R<sup>2</sup>** values remain close to R<sup>2</sup>, confirming that the model is not overfitting and effectively utilizes the predictors.
- **Low Prediction Errors:**
  - **RMSE:** 0.23 (train) and 0.22 (test), indicating minimal error in predictions.
  - **MAE:** 0.18 (train) and 0.17 (test), showing that the average absolute prediction error is very small.
  - **MAPE:** 1.82% (train) and 1.78% (test), suggesting that the model's percentage error is very low.
- **Strong Generalization:**
  - The minimal difference between training and test set performance confirms that the model generalizes well and does not suffer from overfitting.
  - The metrics indicate a reliable and well-optimized linear regression model for predicting the target variable.

## Assumptions of Linear Regression:

These assumptions are essential conditions that should be met before we draw inferences regarding the model estimates or use the model to make a prediction.

For Linear Regression, we need to check if the following assumptions hold:-

1. Linearity
2. Independence
3. Homoscedasticity
4. Normality of error terms
5. No strong Multicollinearity

| Actual Values | Fitted Values | Residuals |
|---------------|---------------|-----------|
| 10.253546     | 10.186066     | 0.06748   |
| 10.062498     | 9.72796       | 0.334538  |
| 10.755624     | 10.706886     | 0.048738  |

|           |           |          |
|-----------|-----------|----------|
| 10.614548 | 10.756445 | -        |
| 10.376144 | 10.557775 | 0.181631 |

Table 1: Comparison between the actual and prediction

## Test for Linearity and Independence:

### Why the test?

- Linearity describes a straight-line relationship between two variables, predictor variables must have a linear relation with the dependent variable.

### How to check linearity?

- Make a plot of fitted values vs residuals. If they don't follow any pattern (the curve is a straight line), then we say the model is linear otherwise model is showing signs of non-linearity.

### How to fix if this assumption is not followed?

- We can try different transformations.

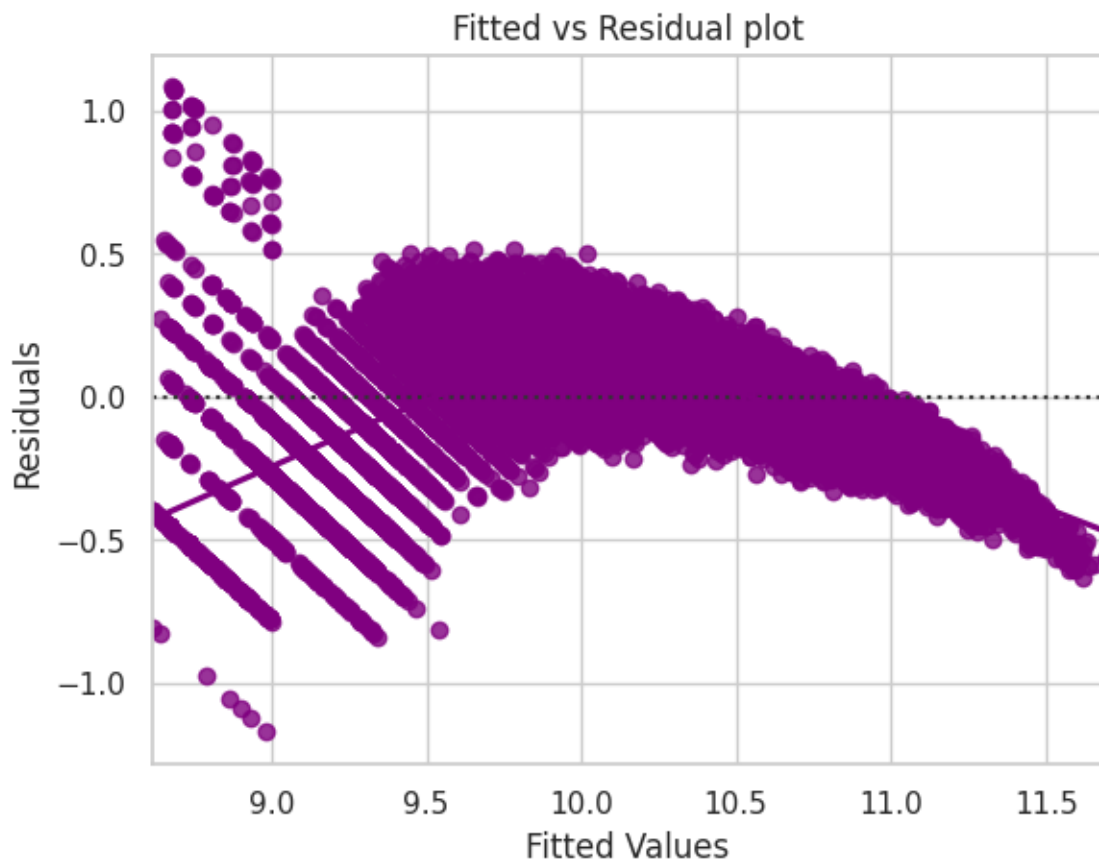


Figure 34: Fitted vs Residuals Plot

- The scatter plot shows the distribution of residuals (errors) vs fitted values (predicted values).
- If there exist any pattern in this plot, we consider it as signs of non-linearity in the data and a pattern means that the model doesn't capture non-linear effects.

- We see no pattern in the plot above. Hence, the assumptions of linearity and independence are satisfied.

### Test for Normality:

What is the test?

- Error terms/residuals should be normally distributed.
- If the error terms are not normally distributed, confidence intervals may become too wide or narrow. Once confidence interval becomes unstable, it leads to difficulty in estimating coefficients based on minimization of least squares.

What does non-normality indicate?

- It suggests that there are a few unusual data points which must be studied closely to make a better model.

How to check the Normality?

- It can be checked via QQ Plot - residuals following normal distribution will make a straight line plot, otherwise not.
- Another test to check for normality is the Shapiro-Wilk test.

How to Make residuals normal?

- We can apply transformations like log, exponential, arcsinh, etc as per our data.

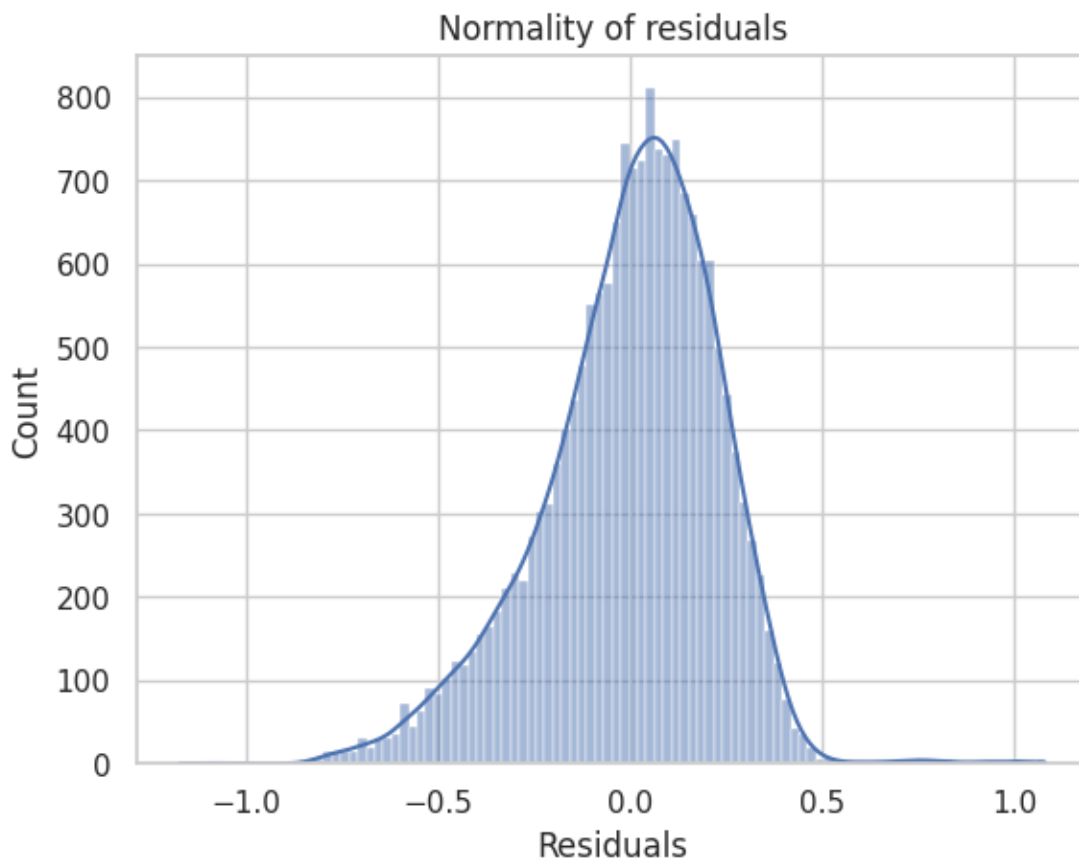


Figure 35: Normality of Residuals

- The residual terms are normally distributed

The Q-Q plot of residuals can be used to visually check the normality assumption. The normal probability plot of residuals should approximately follow a straight line.

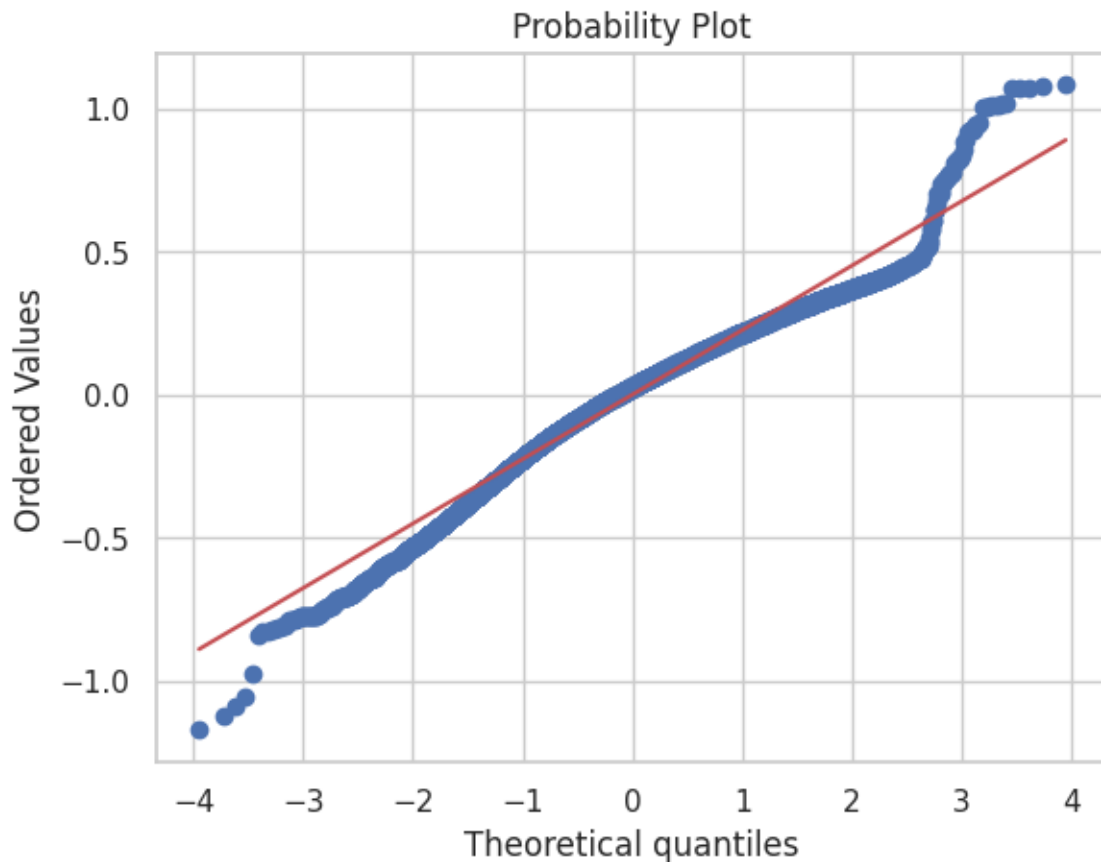


Figure 36: Probability Plot

- Most of the points are lying out of the straight line in QQ plot

The **Shapiro-Wilk** test can also be used for checking the normality. The null and alternate hypotheses of the test are as follows:

- Null hypothesis - Data is normally distributed.
- Alternate hypothesis - Data is not normally distributed.
- ShapiroResult(statistic=0.9756950484397175, pvalue=9.40347720736352e-47)
- Since  $p\text{-value} < 0.05$ , the residuals are not normal as per shapiro test.

### Test for Homoscedasticity:

- Homoscedacity - If the variance of the residuals are symmetrically distributed across the regression line , then the data is said to homoscedastic.
- Heteroscedacity - If the variance is unequal for the residuals across the regression line, then the data is said to be heteroscedastic. In this case the residuals can form an arrow shape or any other non symmetrical shape.

Why the test?

- The presence of non-constant variance in the error terms results in heteroscedasticity. Generally, non-constant variance arises in presence of outliers. How to check if model has Heteroscedasticity?
- Can use the goldfeldquandt test. If we get p-value  $> 0.05$  we can say that the residuals are homoscedastic, otherwise they are heteroscedastic. How to deal with Heteroscedasticity?
- Can be fixed via adding other important features or making transformations. The null and alternate hypotheses of the goldfeldquandt test are as follows:
- `[(F statistic', 1.0243881179463996), ('p-value', 0.12999779120605062)]`

Null hypothesis : Residuals are homoscedastic Alternate hypothesis : Residuals have heteroscedasticity

- Since p-value  $> 0.05$  we can say that the residuals are homoscedastic.

|       | Actual   | Predicted |
|-------|----------|-----------|
| 6824  | 9.683028 | 9.586803  |
| 6673  | 10.33692 | 10.25941  |
| 5822  | 10.16258 | 10.18687  |
| 7829  | 10.41388 | 10.01574  |
| 4075  | 10.0625  | 9.860857  |
| 24688 | 9.60299  | 9.434193  |
| 13129 | 10.45025 | 10.42487  |
| 5146  | 9.60299  | 9.71576   |
| 9958  | 10.98923 | 11.06736  |
| 8130  | 10.16258 | 10.17755  |

Table 2: Comparison between the actual and prediction

- We can observe here that our model has returned pretty good prediction results, and the actual and predicted values are comparable

## Decision Tree Regressor:

Simple Model Performance:

- Training Set Metrics: `{'R2': 1.0, 'Adj. R2': 1.0, 'RMSE': 0.0, 'MAE': 0.0, 'MAPE (%)': 0.0}`
- Test Set Metrics: `{'R2': 0.8923, 'Adj. R2': 0.892, 'RMSE': 0.21, 'MAE': 0.15, 'MAPE (%)': 1.57}`

Observations on the Decision Tree Regression Model:

- The **training set metrics** indicate perfect model performance, with  **$R^2 = 1.0$** , **RMSE = 0.0**, **MAE = 0.0**, and **MAPE = 0.0%**. This suggests that the model has completely memorized the training data, leading to **overfitting**.
- On the **test set**, the model achieves  **$R^2 = 0.8923$** , which is relatively high, indicating good predictive power. However, the drop from perfect training performance to a lower test performance suggests that the model may not generalize well to unseen data.
- The **RMSE (0.21) and MAE (0.15)** on the test set indicate some prediction errors, though they remain within a reasonable range. The **MAPE of 1.57%** shows that the model's percentage error is low.



- Overall, while the Decision Tree performs well on the test set, the extreme overfitting on the training data suggests that **pruning or tuning hyperparameters** (such as max depth, min samples split, or min samples leaf) may be necessary to improve generalization and prevent overfitting.

#### Tuned Decision Tree Model:

- Training Set Metrics: {'R<sup>2</sup>': 0.7093, 'Adj. R<sup>2</sup>': 0.7089, 'RMSE': 0.36, 'MAE': 0.29, 'MAPE (%)': 2.91}
- Test Set Metrics: {'R<sup>2</sup>': 0.6966, 'Adj. R<sup>2</sup>': 0.6957, 'RMSE': 0.36, 'MAE': 0.29, 'MAPE (%)': 2.93}

#### Observations on the Tuned Decision Tree Model:

- The **training set performance** shows an **R<sup>2</sup> of 0.7093**, indicating that the model explains approximately **71% of the variance** in the training data. The **RMSE (0.36) and MAE (0.29)** suggest moderate errors, and the **MAPE of 2.91%** indicates that, on average, predictions deviate by about **2.91%** from actual values.
- The **test set performance** is similar, with an **R<sup>2</sup> of 0.6966**, showing that the model maintains its predictive ability on unseen data. The **RMSE (0.36) and MAE (0.29)** remain consistent, suggesting that the model is **not overfitting** and generalizes well.
- Compared to the **untuned Decision Tree**, which was highly overfitted, the tuned model has **better generalization**, though at the cost of lower training accuracy. This balance suggests that **pruning and hyperparameter tuning** have effectively reduced overfitting while maintaining reasonable predictive power.
- Overall, while the model performs well, further improvements could be achieved by experimenting with **feature selection, ensemble methods (such as Random Forest or Gradient Boosting), or fine-tuning additional hyperparameters** to enhance accuracy.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 1          |
| const                          | 0          |
| years_of_insurance_with_us     | 0          |
| fat_percentage                 | 0          |
| weight_change_in_last_one_year | 0          |
| exercise                       | 0          |
| Alcohol                        | 0          |
| covered_by_any_other_company   | 0          |
| smoking_status                 | 0          |
| bmi                            | 0          |
| avg_glucose_level              | 0          |
| Gender                         | 0          |
| other_major_decs_history       | 0          |
| heart_decs_history             | 0          |
| age                            | 0          |
| daily_avg_steps                | 0          |
| cholesterol_level              | 0          |
| visited_doctor_last_1_year     | 0          |
| Occupation                     | 0          |

|                            |   |
|----------------------------|---|
| adventure_sports           | 0 |
| regular_checkup_lasy_year  | 0 |
| Years_since_last_admission | 0 |

Table 3: Important Variables from Tuned Decision Tree Model

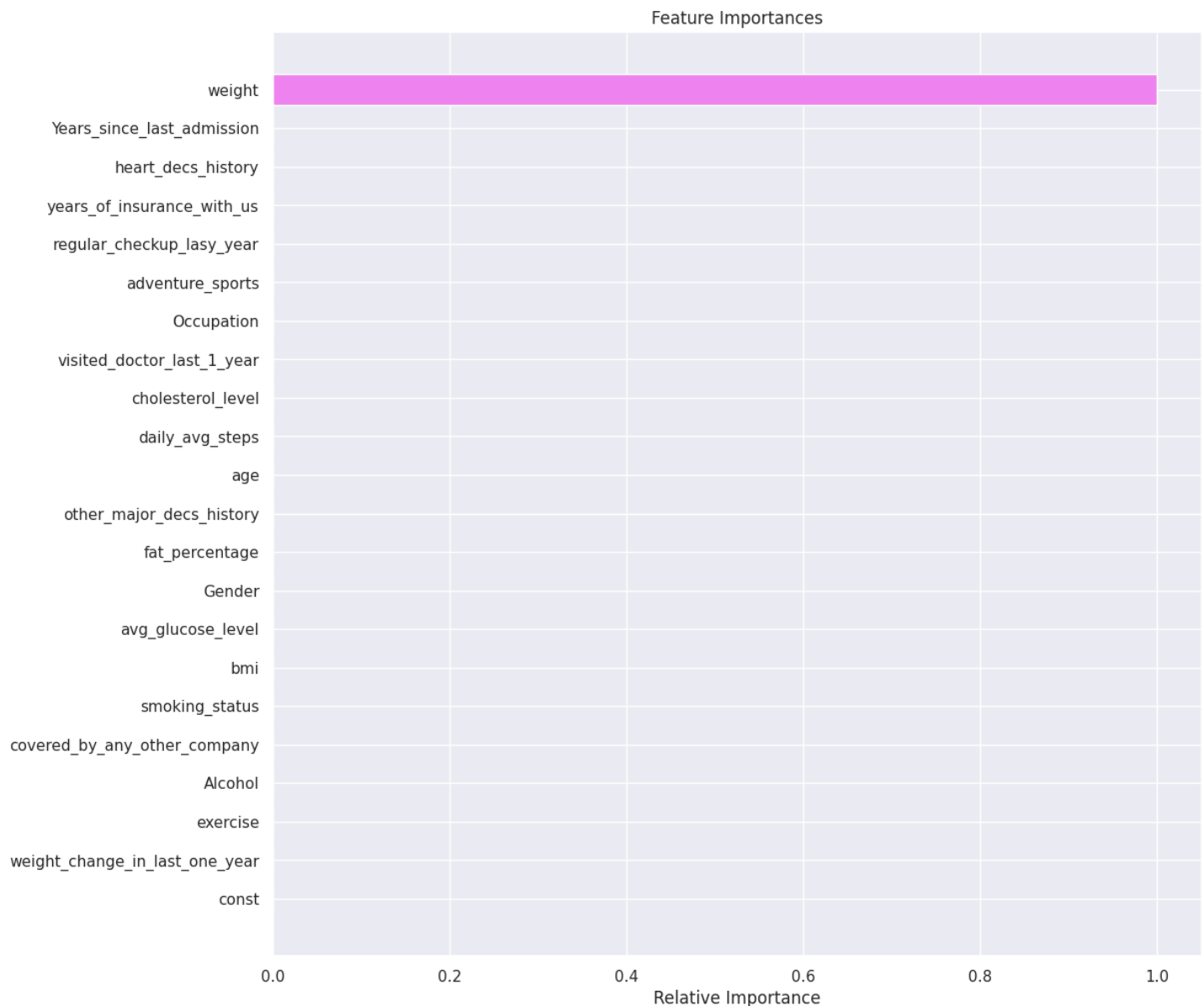


Figure 37: Tuned Bagging Regressor Feature Importance

- The tuned Decision Tree model has assigned importance only to the weight of the customer, indicating that other features were either not influential or were pruned during the tuning process.
- This suggests that, according to the model, body weight is the most significant predictor in determining the target variable. However, relying on a single feature might lead to a biased or overly simplistic model that ignores other potential influencing factors.
- To improve the model, it may be beneficial to re-evaluate feature selection, explore interaction effects between variables, or try alternative models like Random Forest or XGBoost, which consider multiple features and reduce bias.

## Bagging Regressor Model:

### Simple Bagging Regressor Model:

- Training Set Metrics: {'R<sup>2</sup>': 0.99, 'Adj. R<sup>2</sup>': 0.99, 'RMSE': 0.07, 'MAE': 0.04, 'MAPE (%)': 0.46}
- Test Set Metrics: {'R<sup>2</sup>': 0.9404, 'Adj. R<sup>2</sup>': 0.9403, 'RMSE': 0.16, 'MAE': 0.12, 'MAPE (%)': 1.21}

### Observation on Bagging Regressor Performance:

- The **Bagging Regressor shows strong performance** on both the training and test sets, with an **R<sup>2</sup> of 0.99 on training data and 0.9404 on test data**, indicating a **highly predictive model** with minimal loss in generalization.
- The **low RMSE (0.07 on training, 0.16 on test)** and **low MAE (0.04 on training, 0.12 on test)** suggest that the model's predictions are quite accurate, with small deviations from the actual values.
- The **MAPE (%) of 0.46% on training and 1.21% on test** further confirms that the model has **high precision** and is able to generalize well.
- Although there is **some performance drop between training and testing**, the gap is not extreme, indicating that the model is **not heavily overfitting** but still retains strong predictive capability.
- Further improvements could involve **hyperparameter tuning** or **feature selection optimization** to further reduce the slight variance between training and test performance.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.932673   |
| Years_since_last_admission     | 0.011459   |
| weight_change_in_last_one_year | 0.008015   |
| bmi                            | 0.006645   |
| daily_avg_steps                | 0.006518   |
| avg_glucose_level              | 0.006332   |
| age                            | 0.005565   |
| fat_percentage                 | 0.004627   |
| years_of_insurance_with_us     | 0.003513   |
| regular_checkup_lasy_year      | 0.002306   |
| visited_doctor_last_1_year     | 0.002035   |
| smoking_status                 | 0.001735   |
| covered_by_any_other_company   | 0.001695   |
| cholesterol_level              | 0.00169    |
| exercise                       | 0.0013     |
| Alcohol                        | 0.001185   |
| Occupation                     | 0.001031   |
| Gender                         | 0.000627   |
| other_major_decs_history       | 0.000499   |
| heart_decs_history             | 0.000319   |
| adventure_sports               | 0.000232   |
| const                          | 0          |

Table 4: Simple Bagging Regressor Variable Importance

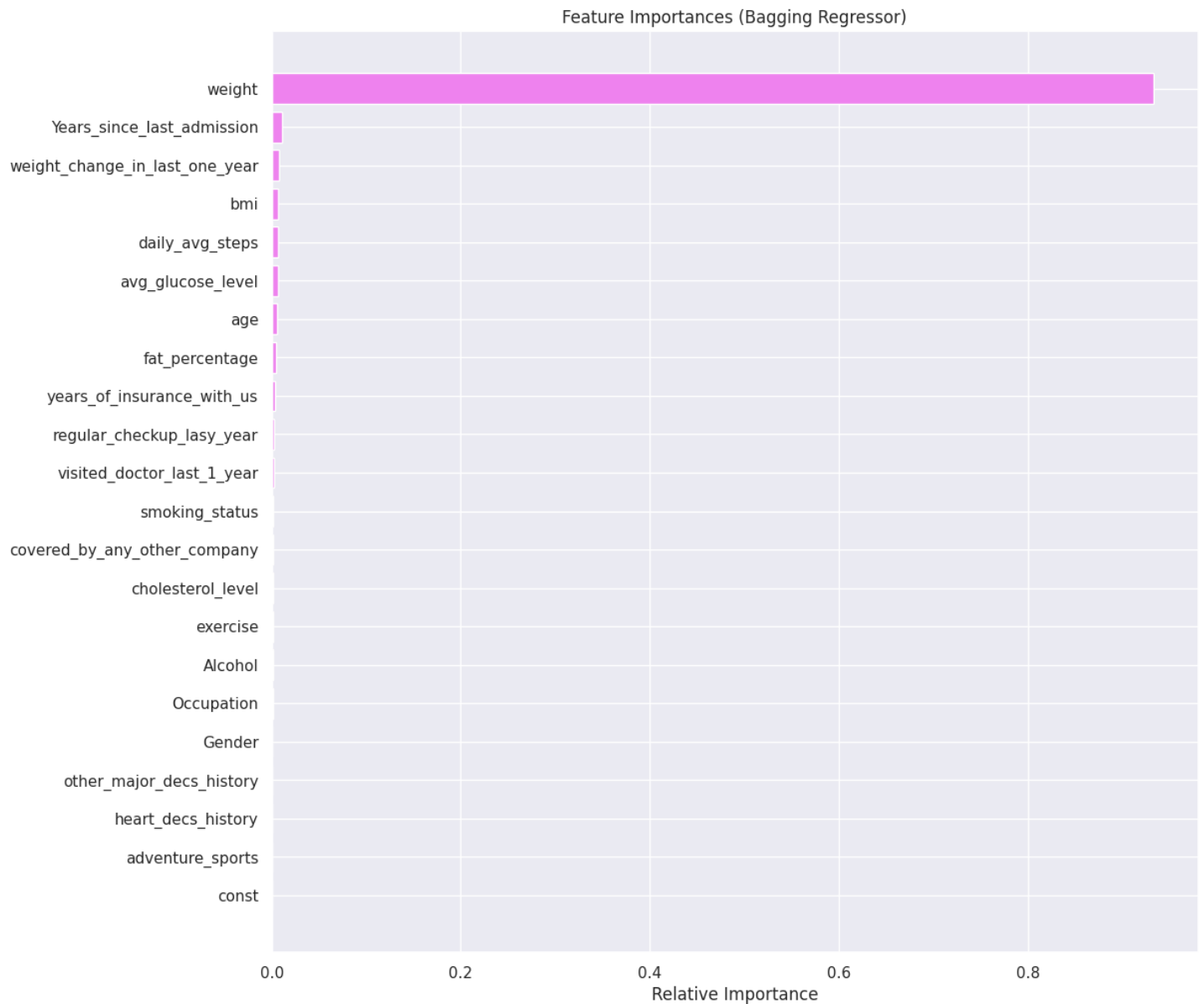


Figure 38: Simple Bagging Regressor Feature Importance

#### Feature Importance Analysis for Bagging Regressor:

- **Weight is the most influential variable**, contributing **93.27%** to the model's predictions, indicating that body weight is the primary determinant of the target variable.
- **Years since last admission (1.15%)** and **weight change in the last year (0.8%)** have minor but notable importance, suggesting that recent health status and weight fluctuations impact the prediction.
- **BMI (0.66%), daily average steps (0.65%), and average glucose level (0.63%)** play a small role, likely reflecting overall health and metabolic condition.
- **Demographic factors like age (0.56%) and fat percentage (0.46%)** have marginal contributions, meaning they provide some predictive power but are not as critical.
- **Lifestyle factors such as smoking status (0.17%), exercise (0.13%), and alcohol consumption (0.12%)** have very little influence, implying that their effect on the outcome is minimal in the model.

- **Insurance-related features like years with the company (0.35%) and other coverage (0.16%)** are not significant predictors.
- **Historical health conditions (heart disease history - 0.03%, major disease history - 0.05%) and adventure sports participation (0.02%)** have negligible contributions.
- The **constant (bias term) has an importance score of zero**, meaning the model relies entirely on feature-based predictions rather than an inherent bias.

#### Tuned Bagging Regressor Model:

- Training Set Metrics: {'R<sup>2</sup>': 0.9912, 'Adj. R<sup>2</sup>': 0.9912, 'RMSE': 0.06, 'MAE': 0.05, 'MAPE (%)': 0.47}
- Test Set Metrics: {'R<sup>2</sup>': 0.9419, 'Adj. R<sup>2</sup>': 0.9417, 'RMSE': 0.16, 'MAE': 0.12, 'MAPE (%)': 1.21}

#### Observations on the Tuned Bagging Regressor Model:

- **Model Performance:** The **training R<sup>2</sup> is 0.9912**, indicating that the model captures nearly all variance in the training data. However, the **test R<sup>2</sup> is 0.9419**, showing a slight performance drop, but the model still generalizes well.
- **Error Metrics:** The **RMSE (0.06 for training, 0.16 for testing)** and **MAE (0.05 for training, 0.12 for testing)** suggest that while the model performs exceptionally well on training data, its performance on the test set is slightly lower, but still strong.
- **Potential Overfitting:** The very high training R<sup>2</sup> suggests that the model might be slightly overfitting. However, since the test R<sup>2</sup> is also high, the model still generalizes well. Further tuning, such as adjusting the number of estimators or regularization, might improve generalizability.
- **MAPE (%) Comparison:** The **mean absolute percentage error (MAPE) remains low (0.47% in training, 1.21% in testing)**, confirming that the model makes accurate predictions with minimal percentage errors.

Overall, the **tuned Bagging Regressor performs slightly better than the untuned version**, maintaining strong generalization with only a minor risk of overfitting.

| Variables                    | Importance |
|------------------------------|------------|
| avg_glucose_level            | 0.116656   |
| heart_decs_history           | 0.096361   |
| bmi                          | 0.084928   |
| Occupation                   | 0.084616   |
| years_of_insurance_with_us   | 0.079693   |
| adventure_sports             | 0.06645    |
| covered_by_any_other_company | 0.065697   |
| visited_doctor_last_1_year   | 0.060329   |
| smoking_status               | 0.057513   |
| cholesterol_level            | 0.050988   |
| Gender                       | 0.042493   |
| exercise                     | 0.041579   |
| const                        | 0.028899   |
| age                          | 0.026643   |

|                           |          |
|---------------------------|----------|
| daily_avg_steps           | 0.025499 |
| regular_checkup_lasy_year | 0.025062 |
| weight                    | 0.022661 |
| other_major_decs_history  | 0.013271 |
| Alcohol                   | 0.010662 |

Table 5: Feature Importance Bagging Regressor

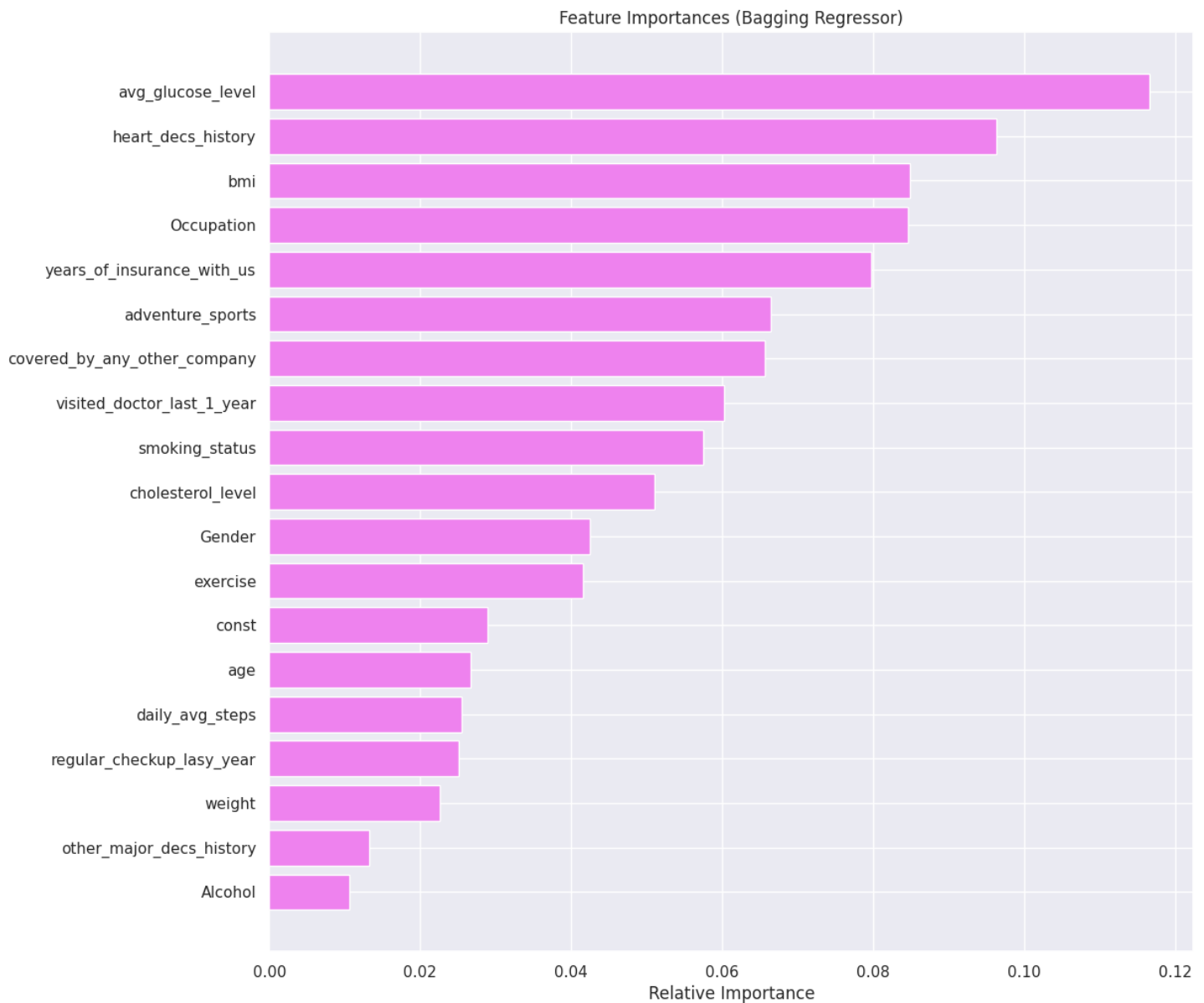


Figure 39: Feature Importance Tuned Bagging Classifier

#### Observations on Feature Importance Tuned Bagging Regressor:

- In earlier models (such as Decision Tree and Bagging Regressor), Weight was the dominant feature. However, in this model, Avg\_glucose\_level and Heart\_disease\_history have taken precedence.
- The feature distribution appears more balanced, meaning the model considers multiple factors rather than relying too heavily on a single variable.
- Unlike the Decision Tree model, where only weight was a key factor, this model distributes importance across health metrics, medical history, and lifestyle factors.

# Random Forest Model:

Simple Random Forest Model:

- ◆ Training Set Metrics: {'R<sup>2</sup>': 0.9926, 'Adj. R<sup>2</sup>': 0.9926, 'RMSE': 0.06, 'MAE': 0.04, 'MAPE (%)': 0.43}
- ◆ Test Set Metrics: {'R<sup>2</sup>': 0.9459, 'Adj. R<sup>2</sup>': 0.9458, 'RMSE': 0.15, 'MAE': 0.11, 'MAPE (%)': 1.15}

Observations on Random Forest Model Performance:

Training Set Performance:

- **R<sup>2</sup> = 0.9926** and **Adj. R<sup>2</sup> = 0.9926** indicate an excellent fit on the training data, showing that the model explains nearly all the variability in the target variable.
- **RMSE (0.06)** and **MAE (0.04)** are very low, suggesting minimal prediction errors.
- The **MAPE (0.43%)** is impressively low, indicating highly accurate percentage error predictions.

Test Set Performance:

- **R<sup>2</sup> = 0.9459** and **Adj. R<sup>2</sup> = 0.9458** highlight strong predictive performance on unseen data, showing the model generalizes well.
- **RMSE (0.15)** and **MAE (0.11)** remain low, confirming good predictive accuracy.
- The **MAPE (1.15%)** indicates strong percentage error control on test data.

Key Observations:

- The model demonstrates excellent performance with minimal signs of overfitting. The slight performance gap between the training and test sets (R<sup>2</sup> drop from 0.9926 to 0.9459) is expected in real-world scenarios.
- Compared to the **Tuned Bagging Regressor** (R<sup>2</sup> = 0.9419), the Random Forest model shows improved test set performance.
- The model balances both low error rates and strong predictive accuracy, making it a robust choice for your regression task.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.933189   |
| Years_since_last_admission     | 0.011246   |
| weight_change_in_last_one_year | 0.008054   |
| daily_avg_steps                | 0.006691   |
| bmi                            | 0.006549   |
| avg_glucose_level              | 0.006213   |
| age                            | 0.005538   |
| fat_percentage                 | 0.004563   |
| years_of_insurance_with_us     | 0.003407   |
| regular_checkup_lasy_year      | 0.00223    |

|                              |          |
|------------------------------|----------|
| visited_doctor_last_1_year   | 0.002044 |
| smoking_status               | 0.001723 |
| covered_by_any_other_company | 0.001653 |
| cholesterol_level            | 0.001647 |
| exercise                     | 0.001337 |
| Alcohol                      | 0.001141 |
| Occupation                   | 0.001012 |
| Gender                       | 0.000618 |
| other_major_decs_history     | 0.000534 |
| heart_decs_history           | 0.000325 |
| adventure_sports             | 0.000285 |

Table 6: Feature Importance Simple Random Forest

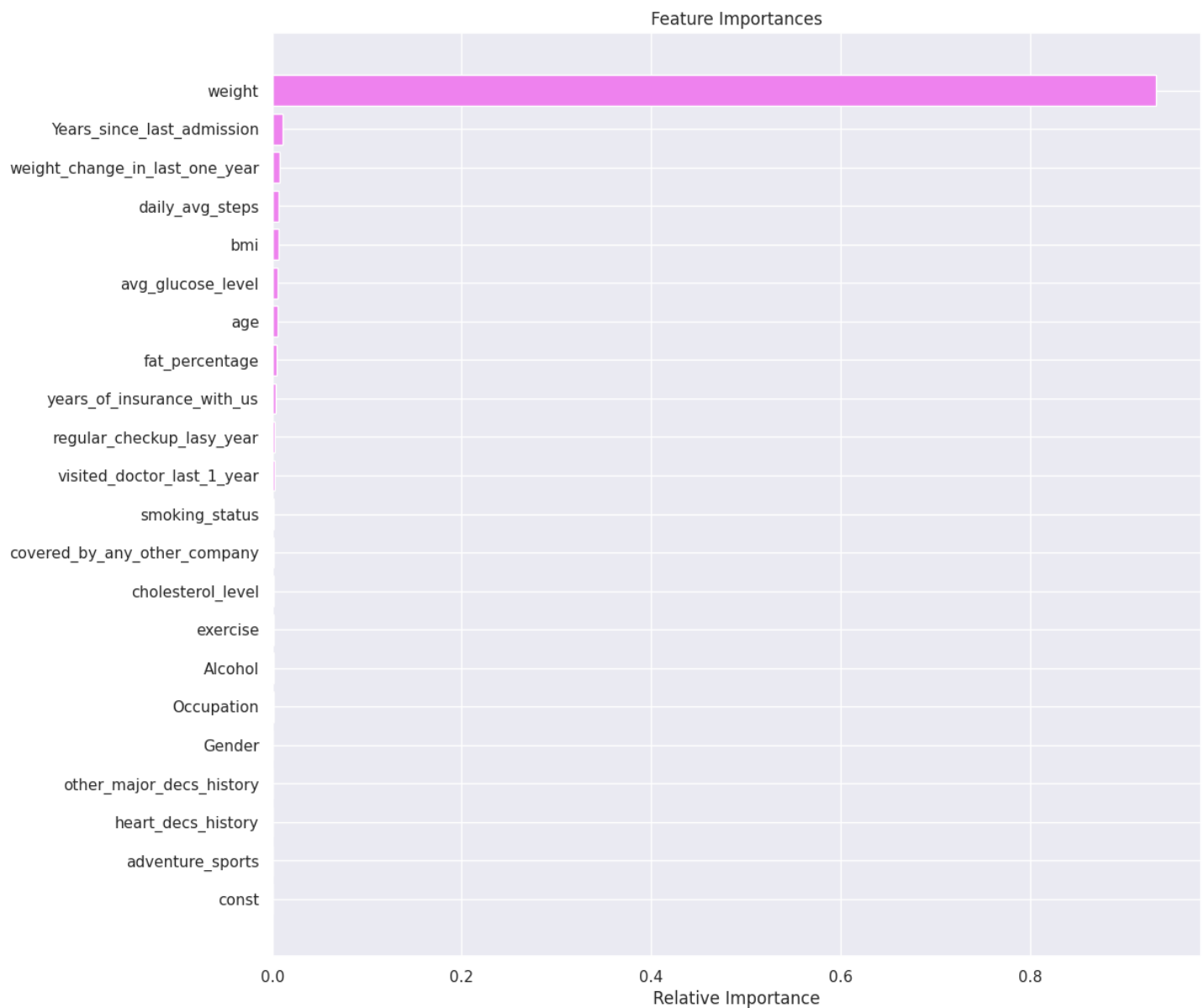


Figure 40: Feature Importance Simple Random Forest

Observations on Feature Importance for the Random Forest Model:

- **Dominating Feature:**



- **Weight** (Importance = **0.933**) is overwhelmingly the most influential factor in the model, indicating it plays a crucial role in predicting the target variable. This dominance suggests the target variable has a strong dependency on body weight.
- **Moderate Influence Features:**
  - Features like **Years\_since\_last\_admission** (0.011), **weight\_change\_in\_last\_one\_year** (0.008), and **daily\_avg\_steps** (0.0067) contribute moderately, implying these factors still have a meaningful impact but are far less significant compared to weight.
- **Low Influence Features:**
  - Variables such as **cholesterol\_level**, **exercise**, and **Alcohol** have minimal importance, indicating limited contribution to the model's predictive power.
- **Negligible Influence Features:**
  - Features like **heart\_decs\_history**, **adventure\_sports**, and **other\_major\_decs\_history** have extremely low importance values, suggesting they have minimal or no impact on the model's performance.

### Key Takeaways:

- The model heavily relies on **weight**, which may indicate potential data imbalance or a strong correlation with the target variable.
- Although some lifestyle factors (e.g., **exercise**, **alcohol**) appear less impactful, they might still provide indirect insights when combined with other variables.
- Consider examining **multicollinearity** or performing further feature engineering to assess if less important variables can be refined or merged to enhance model performance.

### Tuned Random Forest:

- ◆ Training Set Metrics: {'R<sup>2</sup>': 0.9869, 'Adj. R<sup>2</sup>': 0.9869, 'RMSE': 0.08, 'MAE': 0.06, 'MAPE (%)': 0.57}
- ◆ Test Set Metrics: {'R<sup>2</sup>': 0.9474, 'Adj. R<sup>2</sup>': 0.9472, 'RMSE': 0.15, 'MAE': 0.11, 'MAPE (%)': 1.14}

### Observations on Random Forest Model Performance:

#### Training Set Performance:

- The model demonstrates **excellent performance** on the training data with an **R<sup>2</sup> of 0.9869** and minimal errors (RMSE = **0.08**, MAE = **0.06**, MAPE = **0.57%**).
- Such strong metrics suggest the model effectively fits the training data, capturing complex patterns well.

#### Test Set Performance:

- The model maintains **strong generalization** on unseen data with an **R<sup>2</sup> of 0.9474**, indicating that about **94.7% of the variance** in the target variable is explained by the model.
- The error metrics (RMSE = **0.15**, MAE = **0.11**, MAPE = **1.14%**) are reasonably low, showcasing good predictive accuracy.

### Key Observations:

- The slight gap between training and test scores suggests the model has achieved a **balance between accuracy and generalization**, minimizing overfitting.
- Despite the improved performance after tuning, the dominance of **weight** in feature importance may still indicate a potential imbalance in variable influence.
- The model's low **MAPE (%)** highlights its robustness in predicting relative percentage errors, making it particularly suitable for continuous value predictions in your data.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.81706    |
| Years_since_last_admission     | 0.09213    |
| weight_change_in_last_one_year | 0.037473   |
| years_of_insurance_with_us     | 0.007557   |
| daily_avg_steps                | 0.006462   |
| bmi                            | 0.006435   |
| avg_glucose_level              | 0.006193   |
| age                            | 0.005685   |
| fat_percentage                 | 0.004587   |
| regular_checkup_lasy_year      | 0.00336    |
| visited_doctor_last_1_year     | 0.002065   |
| smoking_status                 | 0.001812   |
| covered_by_any_other_company   | 0.001763   |
| cholesterol_level              | 0.001748   |
| exercise                       | 0.001372   |
| Alcohol                        | 0.001243   |
| Occupation                     | 0.001059   |
| Gender                         | 0.000673   |
| other_major_decs_history       | 0.000604   |
| adventure_sports               | 0.000361   |
| heart_decs_history             | 0.000355   |

Table 7: Feature Importance Tuned Random Forest

### Observations on Tuned Random Forest Feature Importance:

#### ◆ Key Variable Dominance:

- The **weight** variable has the highest importance (**0.81706**) by a significant margin, suggesting it is the primary driver of the model's predictions. This heavy reliance could indicate potential **overfitting** or **data imbalance**.

#### ◆ Secondary Influencers:

- **Years since last admission (0.09213)** is the second most influential feature, followed by **weight change in the last one year (0.03747)**. These variables may reflect key health patterns relevant to the prediction target.

#### ◆ Marginal Contributors:

- Variables like **age**, **BMI**, **glucose levels**, and **fat percentage** contribute minimally, yet they still provide some predictive value.

- Features such as **heart disease history**, **adventure sports**, and **gender** have very low importance values, indicating limited influence on model outcomes.

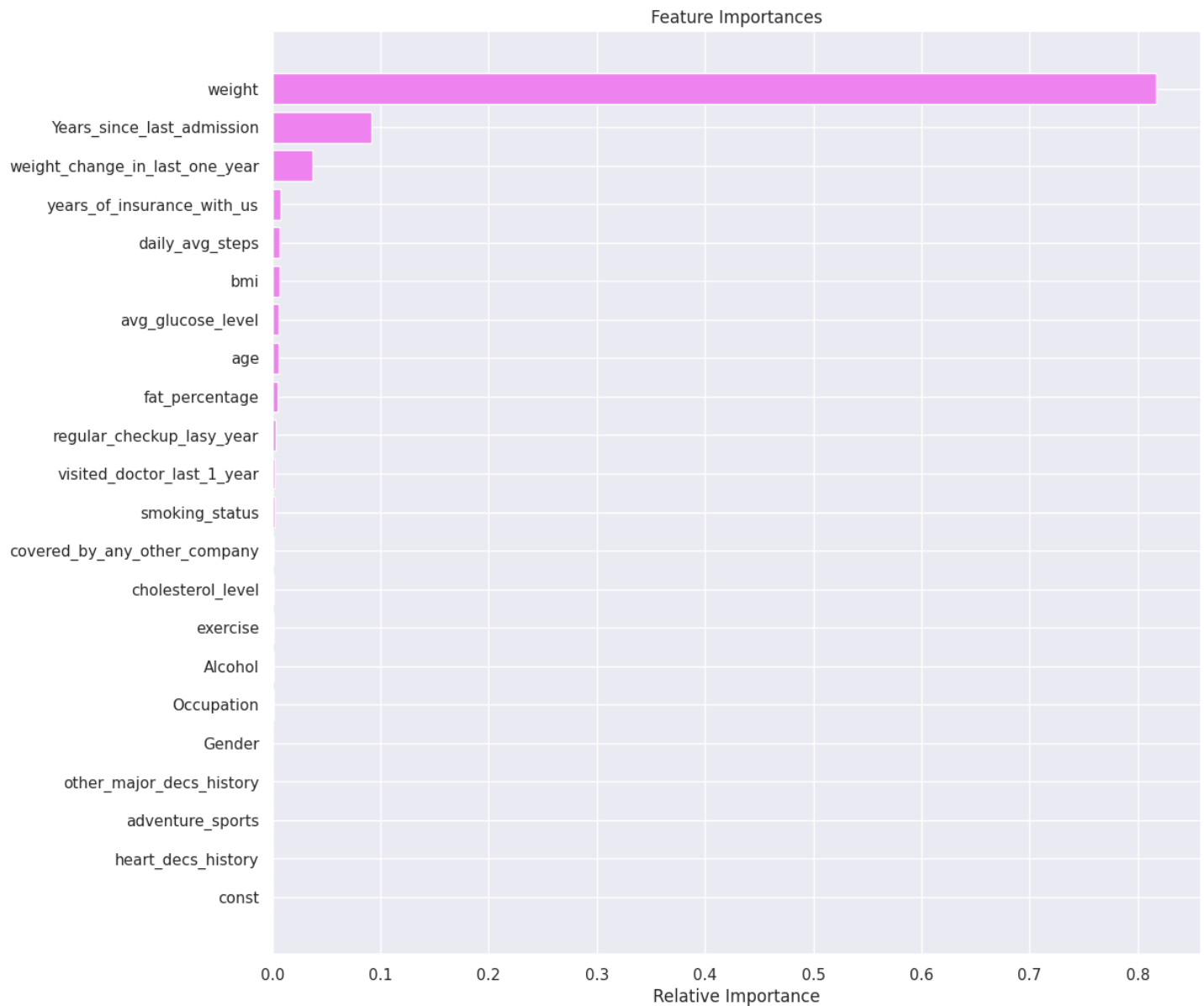


Figure 41: Feature Importance Tuned Random Forest

## Boosting Technique Regression:

### Ada Boosting:

#### Simple Ada Boosting:

- ◆ Training Set Metrics: {'R<sup>2</sup>': 0.9333, 'Adj. R<sup>2</sup>': 0.9332, 'RMSE': 0.17, 'MAE': 0.13, 'MAPE (%)': 1.33}
- ◆ Test Set Metrics: {'R<sup>2</sup>': 0.9333, 'Adj. R<sup>2</sup>': 0.9331, 'RMSE': 0.17, 'MAE': 0.13, 'MAPE (%)': 1.32}

#### Observations on Simple AdaBoost Model Performance:

- Strong Model Performance:**

- The AdaBoost model demonstrates **consistent performance** across both training and test sets with an **R<sup>2</sup> of 0.9333**, indicating that the model explains **93.3%** of the variance in the target variable. This suggests the model has learned the data patterns effectively.
- **Error Metrics Analysis:**
  - The **RMSE** value of **0.17** for both the training and test sets reflects a moderate prediction error, indicating a reasonable fit with stable performance.
  - The **MAE** value of **0.13** and **MAPE** of **~1.3%** further confirm low deviation between predicted and actual values, signifying the model's precision.
- **Balanced Performance:**
  - The similarity between training and test metrics suggests **low risk of overfitting**, implying the model generalizes well to unseen data.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.932807   |
| Years_since_last_admission     | 0.029073   |
| weight_change_in_last_one_year | 0.012859   |
| years_of_insurance_with_us     | 0.010991   |
| regular_checkup_lasy_year      | 0.010874   |
| age                            | 0.003395   |
| bmi                            | 0          |
| fat_percentage                 | 0          |
| exercise                       | 0          |
| Alcohol                        | 0          |
| covered_by_any_other_company   | 0          |
| smoking_status                 | 0          |
| const                          | 0          |
| avg_glucose_level              | 0          |
| other_major_decs_history       | 0          |
| heart_decs_history             | 0          |
| daily_avg_steps                | 0          |
| cholesterol_level              | 0          |
| visited_doctor_last_1_year     | 0          |
| Occupation                     | 0          |
| adventure_sports               | 0          |

*Table 8: Feature Importance Ada Boost*

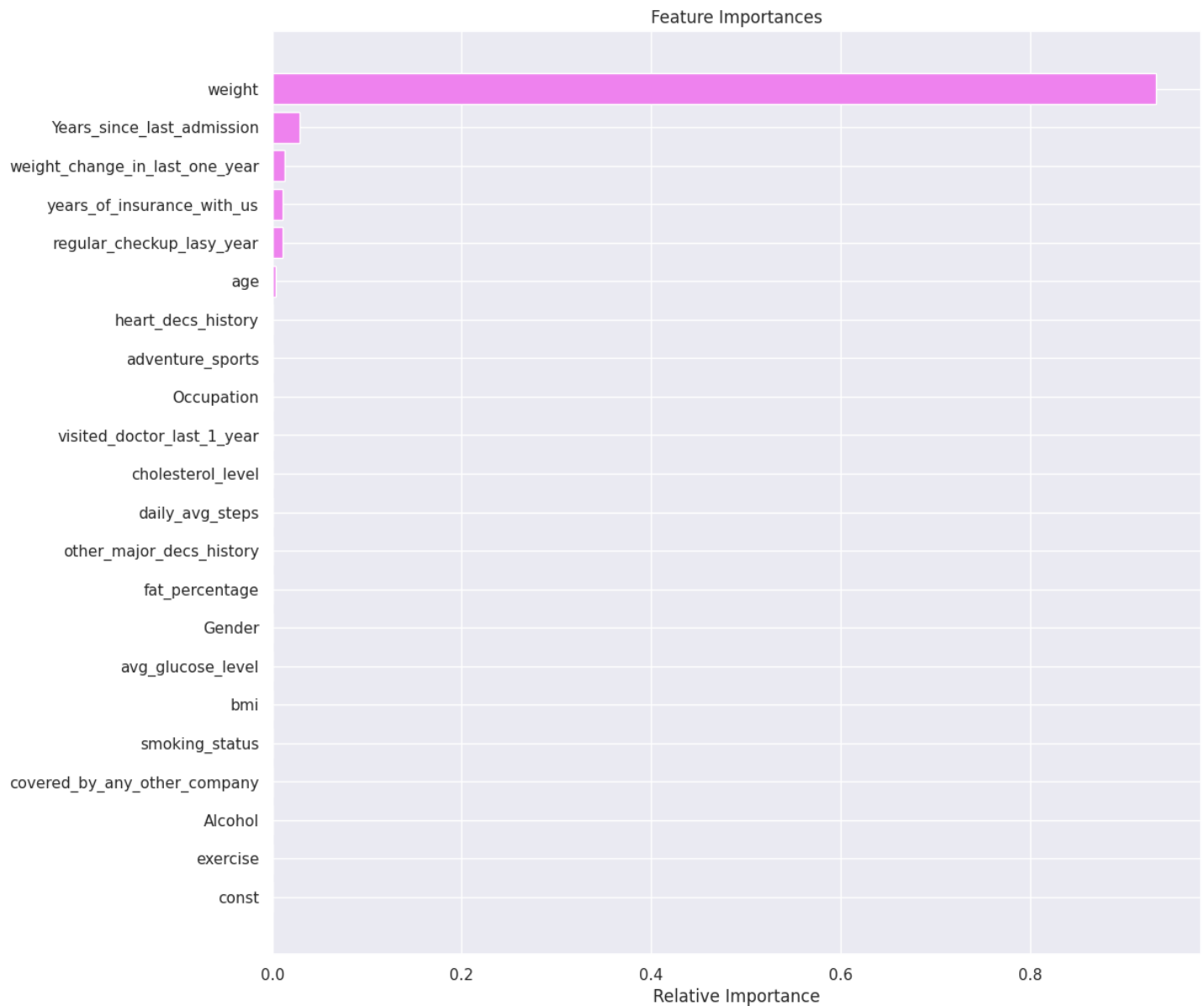


Figure 42: Feature Importance Simple Ada Boosting

### Observations on Simple AdaBoost Feature Importance:

- **Dominance of Key Features:**
  - The AdaBoost model has placed **overwhelming importance** on the **weight** variable (**0.9328**), suggesting that weight is the primary factor driving predictions.
  - Secondary contributors include:
    - **Years since last admission (0.029)**
    - **Weight change in the last one year (0.0129)**
    - **Years of insurance with us (0.011)**
    - **Regular checkup in the last year (0.0109)**
- **Negligible or Zero Importance for Many Features:**

- Several variables, such as **BMI**, **exercise**, **alcohol**, and **cholesterol level**, show **zero importance**, meaning the AdaBoost model deemed them irrelevant for prediction.
- **Potential Overfitting Risk:**
  - The extreme focus on a single feature (**weight**) might indicate an **overfitting issue**, where the model relies too heavily on one dominant factor. This may reduce its ability to generalize well in different scenarios.

Tuned Ada Boosting:

◆ Training Set Metrics: {'R<sup>2</sup>': 0.9323, 'Adj. R<sup>2</sup>': 0.9322, 'RMSE': 0.17, 'MAE': 0.13, 'MAPE (%)': 1.34}

◆ Test Set Metrics: {'R<sup>2</sup>': 0.9322, 'Adj. R<sup>2</sup>': 0.932, 'RMSE': 0.17, 'MAE': 0.13, 'MAPE (%)': 1.33}

#### Observations on Tuned AdaBoost Model Performance:

- **Model Performance Analysis:**
  - The **training** and **test** metrics are very similar, indicating that the model has achieved **good generalization** without significant overfitting or underfitting.
  - The **R<sup>2</sup> values** for both sets (~**0.93**) demonstrate strong predictive accuracy, though slightly lower than some other models like Random Forest or Bagging.
  - **RMSE**, **MAE**, and **MAPE** values are also consistent across both sets, reinforcing stable model behavior.
- **Strengths:**
  - The model maintains **consistent performance** across training and test data.
  - The model effectively balances **bias and variance**, ensuring predictions are stable.
- **Limitations:**
  - Despite stable performance, the model's **R<sup>2</sup> score is slightly lower** than models like **Random Forest** or **Bagging**, suggesting that AdaBoost may have sacrificed some predictive power in exchange for stability.
  - As seen from the feature importance analysis, the model heavily emphasizes a **single dominant feature (weight)**, which may reduce adaptability to unseen data.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.934838   |
| Years_since_last_admission     | 0.032826   |
| weight_change_in_last_one_year | 0.017449   |
| years_of_insurance_with_us     | 0.007629   |
| regular_checkup_lasy_year      | 0.005648   |
| age                            | 0.001609   |
| bmi                            | 0          |
| fat_percentage                 | 0          |
| exercise                       | 0          |

|                              |   |
|------------------------------|---|
| Alcohol                      | 0 |
| covered_by_any_other_company | 0 |
| smoking_status               | 0 |
| const                        | 0 |
| avg_glucose_level            | 0 |
| other_major_decs_history     | 0 |
| heart_decs_history           | 0 |
| daily_avg_steps              | 0 |
| cholesterol_level            | 0 |
| visited_doctor_last_1_year   | 0 |
| Occupation                   | 0 |
| adventure_sports             | 0 |

Table 9: Feature Importance Tunned Ada Boosting

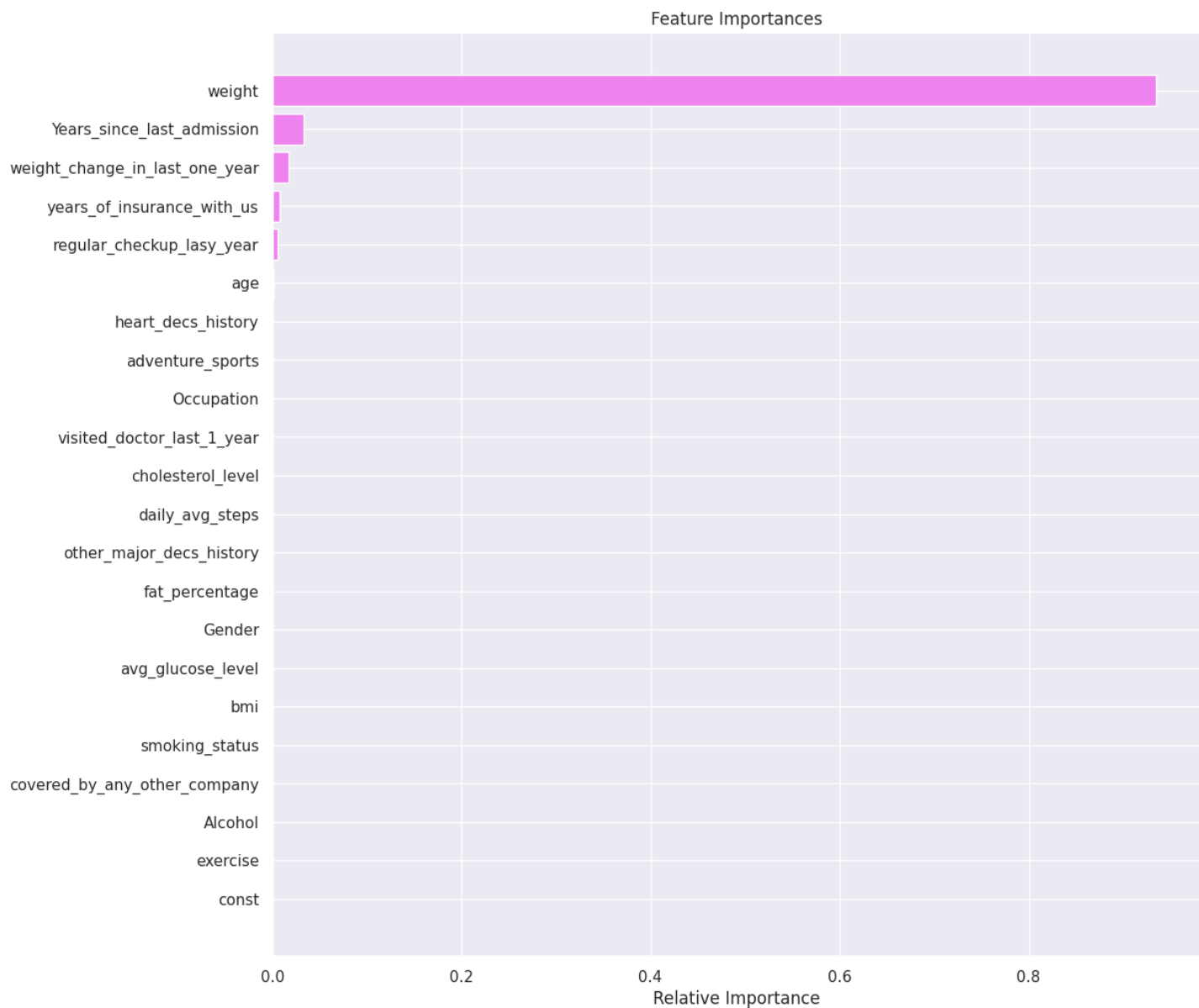


Figure 43: Feature Importance Tunned Ada Boosting

**Observations on Feature Importance (Tuned AdaBoost Model):**

- **Dominant Feature:**
  - The model assigns **extremely high importance to 'weight'** (0.9348), indicating that this variable overwhelmingly drives the model's predictions. This heavy reliance may suggest the model is overly dependent on this feature, potentially limiting its robustness in diverse scenarios.
- **Moderately Important Features:**
  - Features like **'Years\_since\_last\_admission'** (0.0328) and **'weight\_change\_in\_last\_one\_year'** (0.0174) provide some additional predictive value but are significantly less influential than 'weight'.
- **Minimal Contribution Features:**
  - Variables such as **'years\_of\_insurance\_with\_us'**, **'regular\_checkup\_last\_year'**, and **'age'** have minimal impact.
  - **All other features**, including **'bmi'**, **'exercise'**, and **'cholesterol\_level'**, contribute **zero importance** to the model.

## Gradient Boosting:

Simple Gradient Boosting:

- ◆ Training Set Metrics: {'R<sup>2</sup>': 0.95, 'Adj. R<sup>2</sup>': 0.9499, 'RMSE': 0.15, 'MAE': 0.11, 'MAPE (%)': 1.12}
- ◆ Test Set Metrics: {'R<sup>2</sup>': 0.9483, 'Adj. R<sup>2</sup>': 0.9481, 'RMSE': 0.15, 'MAE': 0.11, 'MAPE (%)': 1.14}

## Observations on Gradient Boosting Model Performance:

- **Strong Overall Performance:**
  - The model shows **excellent predictive accuracy** with an **R<sup>2</sup> of 0.95** on the training set and **0.9483** on the test set, indicating minimal overfitting.
  - The **Adjusted R<sup>2</sup>** values (0.9499 for training and 0.9481 for test) confirm that the model generalizes well without relying too heavily on excessive predictors.
- **Error Metrics:**
  - The **RMSE** values of **0.15** for both sets indicate a consistently low prediction error, reinforcing the model's reliability.
  - The **MAE** values of **0.11** highlight the model's strong ability to predict individual data points with minimal deviation.
  - The **MAPE (%)** values (1.12% for training, 1.14% for test) demonstrate that the model maintains excellent predictive accuracy across varying scales.
- **Balanced Model:**
  - The negligible difference between the training and test metrics indicates that the model is well-tuned and avoids overfitting.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.979744   |
| Years_since_last_admission     | 0.011446   |
| weight_change_in_last_one_year | 0.005977   |
| covered_by_any_other_company   | 0.001146   |
| regular_checkup_lasy_year      | 0.000805   |
| years_of_insurance_with_us     | 0.00049    |



|                            |          |
|----------------------------|----------|
| daily_avg_steps            | 0.00012  |
| age                        | 0.00007  |
| avg_glucose_level          | 0.000066 |
| bmi                        | 0.000042 |
| fat_percentage             | 0.000034 |
| exercise                   | 0.000013 |
| adventure_sports           | 0.00001  |
| visited_doctor_last_1_year | 0.000009 |
| other_major_decs_history   | 0.000008 |
| Occupation                 | 0.000006 |
| cholesterol_level          | 0.000006 |
| heart_decs_history         | 0.000005 |
| Alcohol                    | 0.000002 |
| const                      | 0        |
| smoking_status             | 0        |

*Table 10: Feature Importance Simple Gradient Boosting*

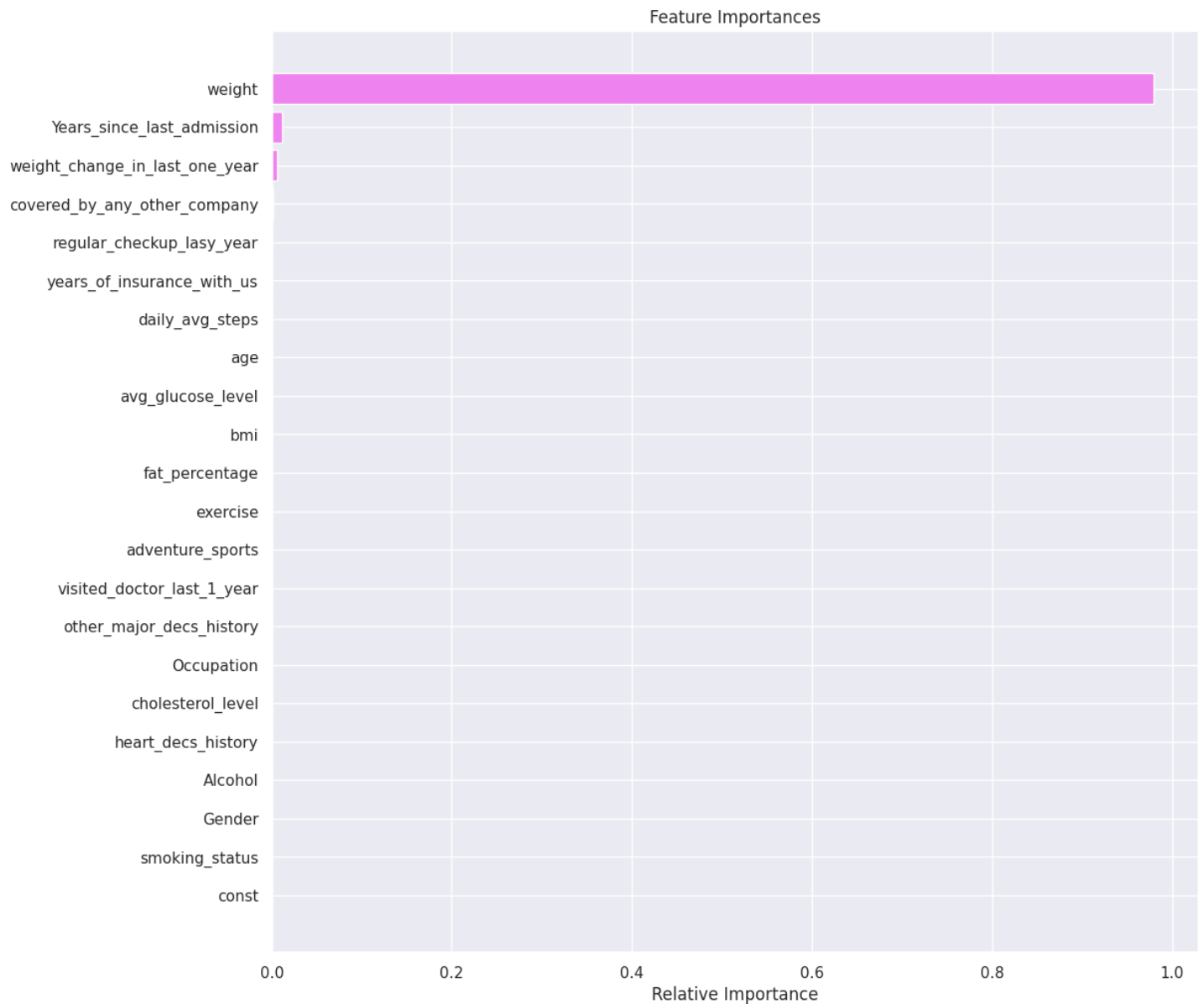


Figure 44: Feature Importance Simple Gradient Boosting

### Observations on Gradient Boosting Feature Importance

- **Key Dominating Feature:**
  - **Weight** is by far the most influential feature, with an importance score of **0.9797**, indicating it plays a major role in the model's predictions. This suggests that weight has a significant impact on the target variable.
- **Moderate Influence Features:**
  - **Years since last admission** (0.0114) and **Weight change in the last one year** (0.0059) contribute moderately to the model's predictive power. These features may relate to health stability and recent lifestyle changes.
- **Minimal Contribution Features:**
  - Features such as **Covered by any other company**, **Regular checkup last year**, and **Years of insurance with us** have minor influence but may still hold some predictive value.
  - Several features like **Alcohol**, **Smoking Status**, and **Gender** have **zero importance**, suggesting they had no measurable impact on the model's performance.

## Tuned Gradient Boosting:

◆ Training Set Metrics: {'R<sup>2</sup>': 0.9557, 'Adj. R<sup>2</sup>': 0.9557, 'RMSE': 0.14, 'MAE': 0.1, 'MAPE (%)': 1.06}

◆ Test Set Metrics: {'R<sup>2</sup>': 0.949, 'Adj. R<sup>2</sup>': 0.9489, 'RMSE': 0.15, 'MAE': 0.11, 'MAPE (%)': 1.13}

## Observations on Tuned Gradient Boosting Model

- **Performance Improvements:**
  - The **Training R<sup>2</sup>** improved to **0.9557**, indicating a stronger fit to the training data.
  - The **Test R<sup>2</sup>** improved slightly to **0.949**, reflecting better generalization.
- **Error Metrics Analysis:**
  - The **RMSE** reduced to **0.14** in the training set and **0.15** in the test set, demonstrating improved accuracy.
  - The **MAE** dropped to **0.10** for training and **0.11** for testing, further confirming better prediction precision.
  - The **MAPE (%)** values improved to **1.06%** in training and **1.13%** in testing, indicating stronger reliability across varying target values.
- **Key Observations:**
  - The reduced gap between the training and test scores suggests the model has effectively balanced **bias** and **variance**, reducing the risk of overfitting.
  - The model maintains a high predictive power while improving generalization, which is crucial for real-world applications.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.902402   |
| Years_since_last_admission     | 0.061199   |
| weight_change_in_last_one_year | 0.021682   |
| years_of_insurance_with_us     | 0.008039   |
| covered_by_any_other_company   | 0.001612   |
| regular_checkup_lasy_year      | 0.001184   |
| daily_avg_steps                | 0.000928   |
| bmi                            | 0.000627   |
| avg_glucose_level              | 0.000569   |
| fat_percentage                 | 0.000528   |
| age                            | 0.000467   |
| cholesterol_level              | 0.000146   |
| visited_doctor_last_1_year     | 0.000123   |
| exercise                       | 0.000117   |
| Occupation                     | 0.0001     |
| other_major_decs_history       | 0.000084   |
| smoking_status                 | 0.000079   |
| Alcohol                        | 0.000055   |
| heart_decs_history             | 0.000028   |
| adventure_sports               | 0.000024   |

Table 11: Feature Importance Tuned Gradient Boosting

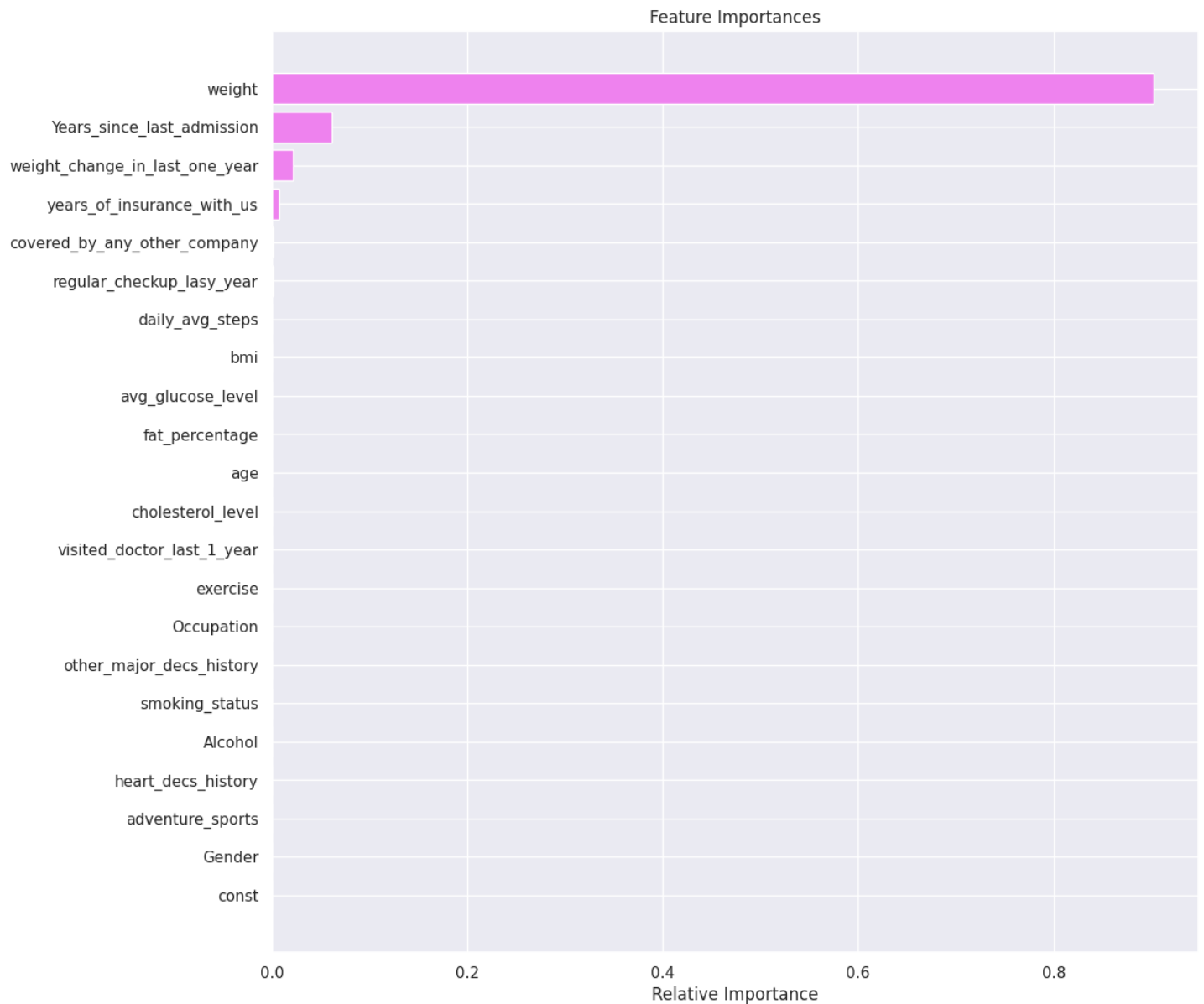


Figure 45: Feature Importance Tuned Gradient Boosting

### Observations on Feature Importance (Tuned Gradient Boosting Model)

- **Dominant Feature:**
  - **Weight** is the most influential feature with an importance score of **0.902**, indicating it has a significant impact on model predictions. This heavy reliance suggests that weight alone contributes disproportionately to the outcome.
- **Moderately Important Features:**
  - **Years Since Last Admission** (0.061) and **Weight Change in Last One Year** (0.021) are the next most impactful variables, indicating some influence on the target variable.
- **Minor Contributions:**
  - Features such as **Years of Insurance with Us**, **Covered by Any Other Company**, and **Regular Checkup Last Year** have minimal influence, each scoring less than **0.01**.
  - Many features like **Alcohol**, **Heart Disease History**, and **Adventure Sports** contribute almost negligibly (close to zero).

## XG Boosting:

### Simple XG Boost:

◆ Training Set Metrics: {'R<sup>2</sup>': 0.978, 'Adj. R<sup>2</sup>': 0.978, 'RMSE': 0.1, 'MAE': 0.07, 'MAPE (%)': 0.75}

◆ Test Set Metrics: {'R<sup>2</sup>': 0.9446, 'Adj. R<sup>2</sup>': 0.9445, 'RMSE': 0.15, 'MAE': 0.11, 'MAPE (%)': 1.17}

### Observation on Simple XG Boosting Model Performance:

- **Training Set Analysis:**

- The model exhibits **excellent performance** on the training data with an **R<sup>2</sup> of 0.978**, indicating it explains **97.8%** of the variance in the target variable.
- **RMSE (0.1)** and **MAE (0.07)** are notably low, suggesting minimal prediction errors.
- The **MAPE of 0.75%** confirms strong predictive accuracy for percentage-based errors.

- **Test Set Analysis:**

- The test performance shows a slight decline with an **R<sup>2</sup> of 0.9446**, still indicating strong generalization.
- **RMSE (0.15)** and **MAE (0.11)** are slightly higher than in training but remain within an acceptable range.
- **MAPE of 1.17%** reflects that the model maintains a high degree of accuracy on unseen data.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.872782   |
| Years_since_last_admission     | 0.026726   |
| weight_change_in_last_one_year | 0.025971   |
| covered_by_any_other_company   | 0.009389   |
| regular_checkup_lasy_year      | 0.005512   |
| Alcohol                        | 0.004625   |
| fat_percentage                 | 0.004608   |
| other_major_decs_history       | 0.004549   |
| exercise                       | 0.004083   |
| cholesterol_level              | 0.004029   |
| bmi                            | 0.00392    |
| daily_avg_steps                | 0.003891   |
| smoking_status                 | 0.003496   |
| avg_glucose_level              | 0.003483   |
| age                            | 0.003449   |
| Gender                         | 0.003356   |
| years_of_insurance_with_us     | 0.003345   |
| Occupation                     | 0.003243   |
| heart_decs_history             | 0.003235   |
| visited_doctor_last_1_year     | 0.003183   |
| adventure_sports               | 0.003126   |

Table 12: Feature Importance Simple XG Boosting

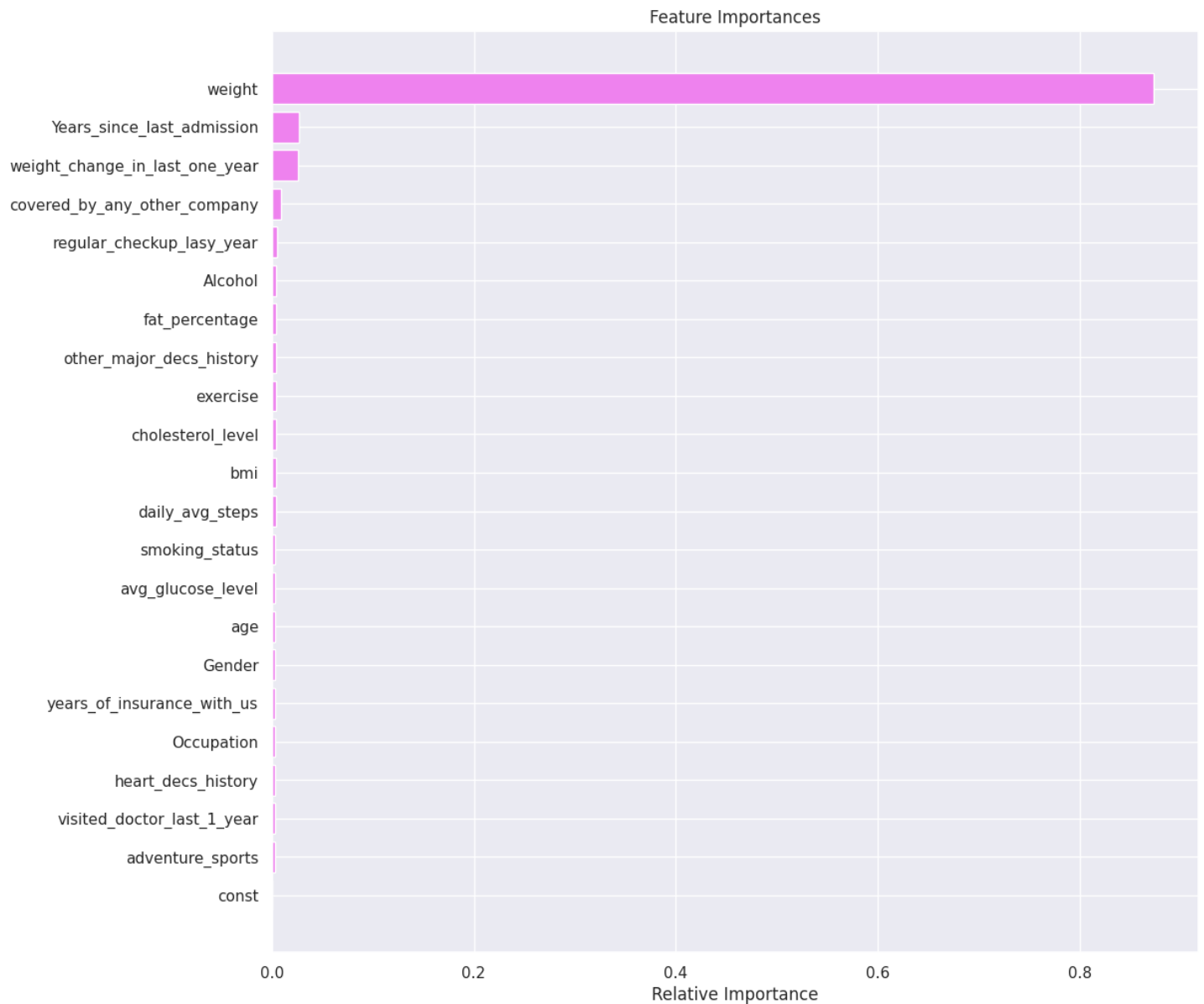


Figure 46: Feature Importance Simple XG Boosting

## Observation on Feature Importance (XG Boosting)

### 1. Dominance of 'Weight':

- The feature **weight** holds the highest importance at **0.8727**, making it the primary driver of model predictions. This suggests that customer weight significantly influences the target variable.

### 2. Secondary Important Features:

- Features like **Years\_since\_last\_admission** (0.0267) and **weight\_change\_in\_last\_one\_year** (0.0260) provide moderate contribution, indicating their role in shaping outcomes but with considerably less influence than weight.

### 3. Minor Contributors:

- Features such as **covered\_by\_any\_other\_company**, **Alcohol**, **cholesterol\_level**, and **smoking\_status** contribute marginally, suggesting they may offer minor predictive value.

#### 4. Even Distribution in Lower Importance Features:

- Features like **adventure\_sports**, **heart\_decs\_history**, and **visited\_doctor\_last\_1\_year** have very low importance scores (around **0.003**), indicating limited impact.

Tuned XG Boost:

◆ Training Set Metrics: {'R<sup>2</sup>': 0.9494, 'Adj. R<sup>2</sup>': 0.9493, 'RMSE': 0.15, 'MAE': 0.11, 'MAPE (%)': 1.14}

◆ Test Set Metrics: {'R<sup>2</sup>': 0.9486, 'Adj. R<sup>2</sup>': 0.9485, 'RMSE': 0.15, 'MAE': 0.11, 'MAPE (%)': 1.14}

#### Observation on Tuned XG Boost Model Performance:

- **Training Set Metrics:**
  - The model demonstrates strong performance with an **R<sup>2</sup> = 0.9494**, indicating it explains approximately **94.94%** of the variance in the training data.
  - Error metrics are low, with an **RMSE = 0.15**, **MAE = 0.11**, and **MAPE = 1.14%**, reflecting solid prediction accuracy.
- **Test Set Metrics:**
  - The model maintains consistent performance on the test set with an **R<sup>2</sup> = 0.9486**, showing minimal overfitting.
  - The error metrics remain stable across train and test sets, with **RMSE**, **MAE**, and **MAPE** values closely aligned.
- **Model Stability:**
  - The negligible performance gap between training and test metrics highlights a **well-generalized model** that effectively balances bias and variance.

| Variables                      | Importance |
|--------------------------------|------------|
| weight                         | 0.914701   |
| weight_change_in_last_one_year | 0.033567   |
| Years_since_last_admission     | 0.027917   |
| regular_checkup_lasy_year      | 0.008142   |
| covered_by_any_other_company   | 0.007921   |
| years_of_insurance_with_us     | 0.007753   |
| avg_glucose_level              | 0          |
| fat_percentage                 | 0          |
| exercise                       | 0          |
| Alcohol                        | 0          |
| smoking_status                 | 0          |
| bmi                            | 0          |

|                            |   |
|----------------------------|---|
| const                      | 0 |
| other_major_decs_history   | 0 |
| heart_decs_history         | 0 |
| age                        | 0 |
| daily_avg_steps            | 0 |
| cholesterol_level          | 0 |
| visited_doctor_last_1_year | 0 |
| Occupation                 | 0 |
| adventure_sports           | 0 |
| Gender                     | 0 |

Table 13: Feature Importance Tuned XG Boosting

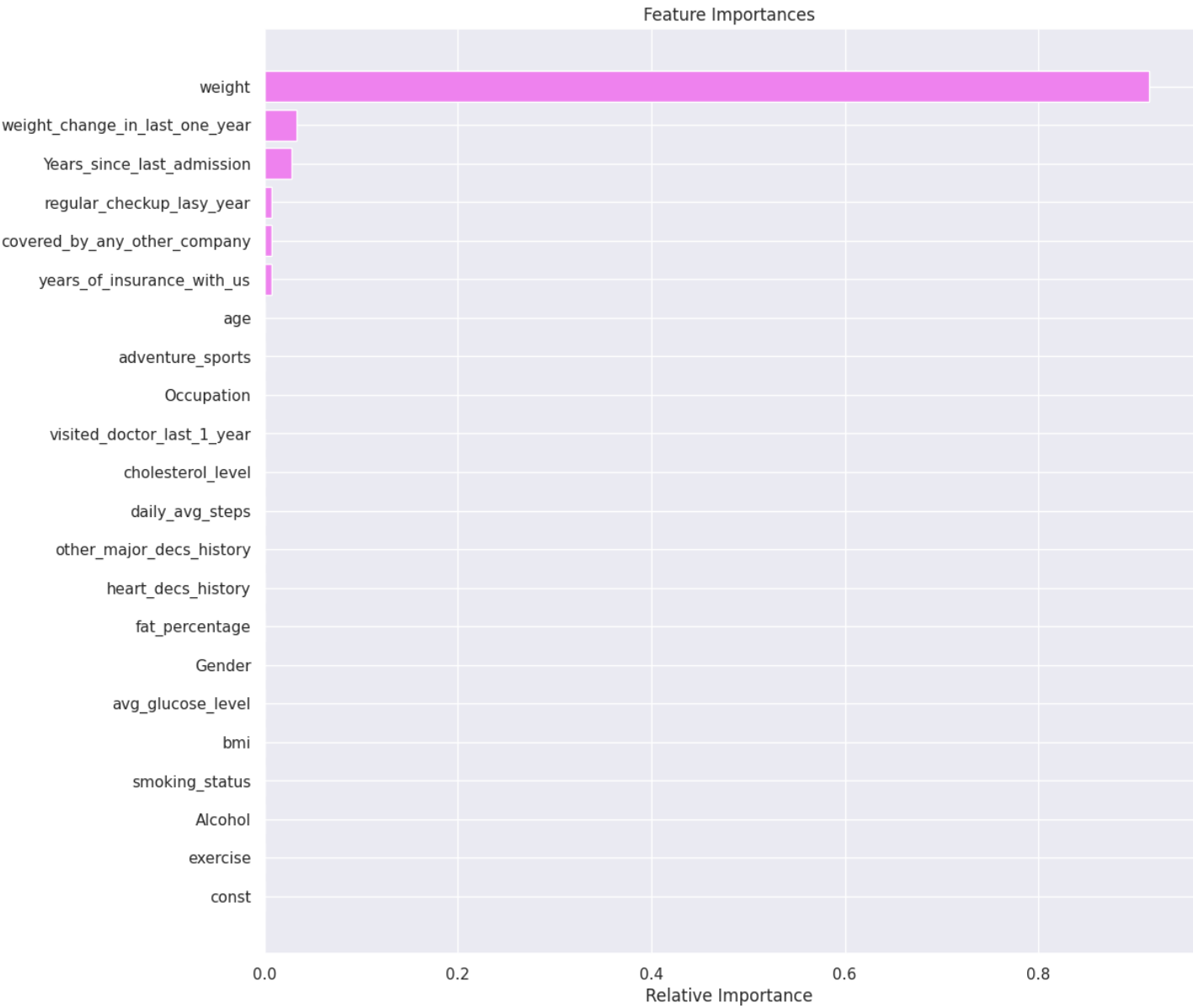


Figure 47: Feature Importance Tuned XG Boosting

Key Observations:

- Dominating Variable:



- **weight** has an overwhelming importance score of **0.9147**, indicating that this single feature drives most of the model's predictions.
- Such heavy reliance on one feature suggests potential data imbalance, model bias, or overfitting.
- **Moderate Importance Features:**
  - **weight\_change\_in\_last\_one\_year** (0.0336)
  - **Years\_since\_last\_admission** (0.0279)

These features contribute some predictive value but are significantly less impactful than weight.
- **Low/Zero Importance Features:**
  - Several meaningful variables like **avg\_glucose\_level**, **bmi**, **exercise**, and **smoking\_status** have **zero importance**, which is unexpected given their potential relevance in healthcare data

## Model Comparison:

|                            | R <sup>2</sup> | Adj. R <sup>2</sup> | RMSE | MAE  | MAPE(%) |
|----------------------------|----------------|---------------------|------|------|---------|
| Models                     | Train Results  |                     |      |      |         |
| Linear Regression train    | 0.8803         | 0.8803              | 0.23 | 0.18 | 1.82    |
| Dtree train                | 1              | 1                   | 0    | 0    | 0       |
| Tuned Dtree train          | 0.7093         | 0.7089              | 0.36 | 0.29 | 2.91    |
| Bagging train              | 0.99           | 0.99                | 0.07 | 0.04 | 0.46    |
| Tuned Bagging train        | 0.9912         | 0.9912              | 0.06 | 0.05 | 0.47    |
| RF train                   | 0.9926         | 0.9926              | 0.06 | 0.04 | 0.43    |
| Tuned RF train             | 0.9869         | 0.9869              | 0.08 | 0.06 | 0.57    |
| Ada Boost train            | 0.9333         | 0.9332              | 0.17 | 0.13 | 1.33    |
| Tuned Ada Boost train      | 0.9323         | 0.9322              | 0.17 | 0.13 | 1.34    |
| Gradient Boost train       | 0.95           | 0.9499              | 0.15 | 0.11 | 1.12    |
| Tuned Gradient Boost train | 0.9557         | 0.9557              | 0.14 | 0.1  | 1.06    |
| XGBoost train              | 0.978          | 0.978               | 0.1  | 0.07 | 0.75    |
| Tuned XGBoost train        | 0.9494         | 0.9493              | 0.15 | 0.11 | 1.14    |
|                            | Test Results   |                     |      |      |         |
| Linear Regression test     | 0.8832         | 0.883               | 0.22 | 0.17 | 1.78    |
| DTree test                 | 0.8923         | 0.892               | 0.21 | 0.15 | 1.57    |
| Tuned DTree test           | 0.6966         | 0.6957              | 0.36 | 0.29 | 2.93    |
| Bagging test               | 0.9404         | 0.9403              | 0.16 | 0.12 | 1.21    |
| Tuned Bagging test         | 0.9419         | 0.9417              | 0.16 | 0.12 | 1.21    |
| RF test                    | 0.9459         | 0.9458              | 0.15 | 0.11 | 1.15    |
| Tuned RF test              | 0.9474         | 0.9472              | 0.15 | 0.11 | 1.14    |
| Ada Boost test             | 0.9333         | 0.9331              | 0.17 | 0.13 | 1.32    |
| Tuned Ada Boost test       | 0.9322         | 0.932               | 0.17 | 0.13 | 1.33    |
| Gradient Boost test        | 0.9483         | 0.9481              | 0.15 | 0.11 | 1.14    |
| Tuned Gradient Boost test  | 0.949          | 0.9489              | 0.15 | 0.11 | 1.13    |
| XGBoost test               | 0.9446         | 0.9445              | 0.15 | 0.11 | 1.17    |

|                    |        |        |      |      |      |
|--------------------|--------|--------|------|------|------|
| Tuned XGBoost test | 0.9486 | 0.9485 | 0.15 | 0.11 | 1.14 |
|--------------------|--------|--------|------|------|------|

Table 14: Model Comparison

- **Tuned Bagging Regressor** uses all variables effectively, which might be appealing. But the **Tuned Bagging Regressor has the highest train  $R^2$  (0.9912)**, but its test  $R^2$  (0.9419) is the lowest among the models. Which suggests it **overfits** the training data and generalizes less effectively to unseen data.
- **Tuned Gradient Boost and Tuned XG Boost** remove less relevant features, leading to more efficient models.

**Tuned Gradient Boosting** is the best choice because:

- Highest Test  $R^2$  (**0.9490**) → Best generalization.
- Lowest Test **MAPE (1.13%)** → More reliable predictions.
- Balances feature selection and performance well.
- Lower overfitting risk compared to Tuned Bagging Regressor.

## Business Insights & Recommendations:

### Key Takeaways from the Model Performance

**Best Model Identified: Tuned Gradient Boosting ( $R^2 = 0.9490$ , MAPE = 1.13%)**

### Key Factors Affecting Insurance Cost:

- **Weight & BMI:** Strongest predictor; obesity increases medical expenses.
- **Years since Last Admission:** More hospital visits = higher insurance costs.
- **Exercise & Smoking Status:** Direct impact on health and insurance pricing.
- **Model Generalization:** The **Tuned Bagging Regressor overfits**, making it less reliable for real-world predictions.

### Business Insights

#### Healthier Lifestyle = Lower Insurance Cost

- **Weight** is the **most influential variable** in determining insurance cost.
- **Exercise & Diet Control (cholesterol, glucose levels)** significantly reduce the risk of high medical expenses.
- **Regular Checkups** help in early diagnosis, indirectly reducing long-term costs.

#### High-Risk Individuals Cost More to Insure

- **Smoking, alcohol consumption, and chronic conditions** lead to higher medical expenses.
- **People with previous major health issues** have a significantly higher insurance cost.
- **Longer hospital stays and frequent doctor visits** indicate greater risk.

### Insurance Tenure & Risk Management

- **Longer policyholders tend to be lower risk** → Encourage long-term customer retention.

- **Individuals switching providers frequently** might indicate undisclosed pre-existing conditions.

## **Recommendations for Insurance Companies**

### **1. Offer Incentives for a Healthier Lifestyle**

- Introduce **discounts for policyholders with a healthy BMI & active lifestyle** (e.g., Fitbit tracking).
- Reward **smoking cessation programs** with premium reductions.
- Provide **free annual health checkups** to encourage early disease detection.

### **2. Personalized & Dynamic Pricing Models**

- Use **AI-driven risk scoring** to adjust premiums dynamically based on health behavior.
- Encourage **wearable health device integration** to track physical activity & adjust premiums.
- Offer **customized plans** based on pre-existing conditions & lifestyle choices.

### **3. Expand Preventive Healthcare Programs**

- Partner with **gyms, dieticians, and mental health programs** to reduce future insurance claims.
- Provide **discounted health screenings** for high-risk individuals to prevent costly treatments later.
- Promote **telemedicine services** for minor issues to reduce hospital visits.

### **4. Fraud Prevention & Risk Reduction**

- Use **machine learning to detect patterns of high-risk policyholders** (frequent claims, hospital visits).
- Monitor **sudden weight gain/loss trends** which might indicate hidden health conditions.

## **Final Thought: How This Impacts Business Strategy**

- Insurance companies can **reduce risk & optimize costs** using AI-based pricing.
- Policyholders benefit from **incentives for healthy living**, making insurance more affordable.
- Early intervention **lowers long-term claim payouts**, improving overall profitability.